Autodesk Data Management Server 2009

Implementation Guide



Part No. 463A1-050000-PM01A January 2008

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Published by:
Autodesk, Inc.
111 McInnis Parkway
San Rafael, CA 94903, USA

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Introduction and System Requirements

This book is intended to help users who have a basic understanding of Microsoft® Windows® to successfully install and implement Autodesk® Vault or Productstream® in a Microsoft Windows network environment. We assume you understand Windows® navigation as well as installing and uninstalling software in a Windows environment

In the following chapters the best practices for implementing the Autodesk® data management server and the recommended methods for deploying the data management server will be explained. Although these are not the only methods for deploying the software, they are the most robust and easiest to maintain.

Use this guide as a starting point. You can always change the configuration later if your business demands it.

NOTE We recommend that you contact your reseller to perform a professional implementation of the Autodesk data management server if you have any questions about this document.

Book Organization

This book addresses the main topics for successfully implementing the document management server. The first topic is a brief understanding of the server components the second topic is the planning and installation of the server and clients. The last topic is maintenance and trouble shooting.

Conventions Used In This Book

The following table illustrates different methods used in this book to make the text easier to read and understand.

Convention	Example
Files or Paths	*C:\Program Files*
World Wide Web Links	*www.autodesk.com*
Important Information	Note Additional information
Program or Script Code	`@ECHO OFF`
User Prompts	`User Name`

Support

Though this book addresses the basic implementation of the data management server for most systems, there are possible combinations of different software and /or hardware interactions not listed. For more information you can visit the links below:

- *www.autodesk.com/vault*
- *www.autodesk.com/discussiongroup-vault*
- *www.autodesk.com/vault-support*
- *www.autodesk.com/productstream*
- *www.autodesk.com/discussiongroup-productstream*
- *www.autodesk.com/productstream-support*
- *www.autodesk.com/vault-whitepapers*

If you are an Autodesk Subscription member you can log support requests through the subscription portal at *www.autodesk.com/subscriptionlogin*

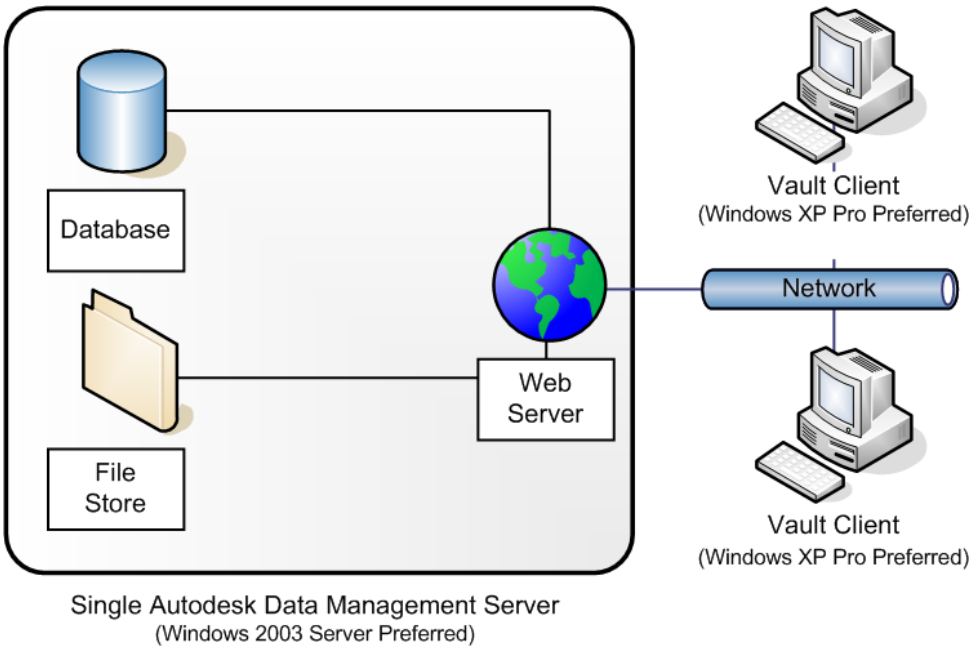
Key Terms

Term	Definition
Data management server	The server used in conjunction with Autodesk Vault, Autodesk Productstream, and Content Center for storing files and version data.
Autodesk web server	A Web server that can be used when one computer will host both the Autodesk data management server and vault client in a stand alone environment, such as an evaluation.
Clients	Applications that access the Autodesk data management server. These include stand alone applications such as Autodesk Vault Explorer and Autodesk Productstream as well as integrated application add-ins such as Autodesk Vault for Inventor, or Autodesk Vault for Microsoft® Office.
Data management	A means to organize and track files and design modifications through the design process.
Autodesk Vault Explorer	The standalone client for Autodesk Vault used to organize information in the vault. This interface can be used to manage any Microsoft Windows file in Autodesk Vault
Microsoft® Internet Information Services (IIS)	A Microsoft® Web server necessary for doing a full implementation of the Autodesk data management server.
Vault	The logical combination of a vault database and a vault file store that contains all of the information managed by Autodesk Vault.

Term	Definition
	Each installation of Autodesk Vault can maintain several independent vaults. A single client can connect to several different vaults, but not simultaneously.
ADMS Console	An application that performs vault system administrative tasks, such as data back up and restore.
Web Server	A server process running on the Autodesk data management server that responds to requests from vault clients.
Web service	An Application Programming Interface for a client to communicate with a web server application, such as the Autodesk data management server.

Understanding Vault Architecture

Two primary components make up the data management server and vault clients. The following figure illustrates the basic configuration of the components.



Autodesk Data Management Server

Autodesk data management server is the server that works in conjunction with Autodesk data management solutions: Autodesk Vault, Autodesk Productstream, and Content Center libraries. It tracks all of the files, Bill of Material data, and properties while managing users and permissions. There are three key components to the server:

- Web server
- Database server
- File store

The server contains a single database instance that hosts both vaults and Content Center libraries.

Web Server

There are two different Web servers supported by the Autodesk data management server:

- Microsoft Internet Information Services (IIS)
- Autodesk web server

The Autodesk web server is a light-weight Web server intended for a single-user environment. The web server only works for configurations where the entire data management environment is installed on a single computer, that is, server and clients with CAD software on the same machine. AWS is also used for accessing Content Center or if you are a stand alone Inventor user working in a non-collaborative environment.

NOTE If IIS is not installed on the computer that will host the server and you require more than a single-user installation, you must install IIS before you install Autodesk data management server.

Autodesk data management server requires IIS for anything more than a single-user installation. Before the server is installed, a check is performed to verify the presence of IIS. If IIS is not detected, it must be installed prior to installing the server.

NOTE You may need the original Microsoft Windows installation CD to complete the IIS installation.

The Autodesk data management server is actually a Web application that uses services to manage the communication between the clients and the server via Web services. The interaction between the clients and the server is similar to the way a web browser communicates with a site on the Internet. The Web services use standard HTTP methods, communicating over the Web server port 80 by default. Microsoft IIS is required for multiple-user configurations.

Database Server

The database server keeps track of all the relationships between data. It acts like a library card catalog, containing indexes and pointers of where to find related information. By default, the Autodesk data management server installs Microsoft® SQL Server 2005 Express. Microsoft SQL Server 2005 Express can

be upgraded any time after the Autodesk data management server components have been installed. The SQL server can be upgraded to:

- Microsoft SQL Server 2005 Enterprise
- Microsoft SQL Server 2005 Workgroup
- Microsoft SQL Server 2005 Standard

File Store

The file store and the database server work with each other. The database provides the index information for where files are located and the file store is the secure location on the server where the files are actually located.

Autodesk Vault Clients

A vault client is any stand alone application or integrated add-in that connects to the Autodesk data management server to access files and perform vault operations.

An example is the Inventor add-in. When data is added to a vault using the Inventor add-in, the add-in preserves all of the complex relationships that are created by assemblies, drawings, presentations, and other files.

Autodesk Vault Explorer

Vault Explorer is a general purpose application for interacting with a vault on the Autodesk data management server. You can browse the complete vault structure, add any file to the vault, and perform most other file-based operations, depending on your level of permission.

Autodesk Vault Add-ins for Design Applications

Add-in clients provide basic vault functions within the environment of a parent application. Add-ins maintain application-specific data relationships when adding files to a vault.

There are add-ins available for Autodesk and non-Autodesk design applications.

NOTE As a general rule, if an integrated client is available for a particular application, managing files using that client minimizes loss of data, such as the assembly relationships. We recommend that you use integrated clients whenever possible.

Microsoft Office Add-in

The Microsoft Office add-in performs basic vault functions on documents, spreadsheets, and other non-CAD data within any of the Microsoft Office applications: Microsoft® Word, Excel®, and Power Point®.

Content Center

Content Center provides access to Autodesk Inventor Content Center libraries. Content Center libraries are stored on the Autodesk data management server.

Autodesk Data Management Utilities

The Autodesk data management server utilities are used to perform maintenance, diagnostics, and batch file uploading.

Server Console

The server console includes tools for maintaining users, vaults, and Content Center libraries. Use the console to purge out-dated versions from a vault. You can schedule the server console to create routine back ups of vault data. The console is also used to maintain replicated sites in a multi-site configuration.

Autodesk Server Diagnostic Tool

The Autodesk Server Diagnostics Tool performs a set of checks to identify possible Autodesk server issues and gathers system information to help diagnose the problem.

Autodesk Autoloader

Autodesk® Autoloader uploads Autodesk Inventor projects, Inventor files, and AutoCAD® *.dwg* files to a vault located on an Autodesk data management server. Use Autoloader to move large Inventor projects into a vault or to bulk upload large numbers of *.dwg* files. Legacy Inventor projects from Inventor 5.3 and newer can be uploaded using Autoloader.

Installation Requirements

Previous versions of Autodesk data management server installed Microsoft® SQL Desktop Edition (MSDE) by default. If an existing version of MSDE or the full SQL server is detected during installation, that SQL server may be used by Autodesk data management server. If no previous version of a SQL server is detected or if you are not connecting to an existing remote SQL instance, Microsoft® SQL Server 2005 Express is installed by default. When using MSDE, Autodesk data management server supports approximately 10 simultaneous vault clients. Additional users can be supported using either Microsoft SQL Server 2005 Express or the full version of Microsoft SQL Server 2005. You must purchase the full Microsoft SQL Server 2005 software separately from Microsoft® or a Microsoft reseller. Install Autodesk data management server with the default version first and then determine the need to upgrade to the full SQL server.

Visit *www.autodesk.com/vault* for more information about system requirements, or the Autodesk® Installation online Help.

NOTE See SQL Server Upgrade on page 20 if your system exceeds 10 simultaneous clients and you are currently using MSDE.

If a compatible local SQL database is detected during installation, you can chose to connect to that instance. You are prompted to migrate the data once the installation is complete.

NOTE If a database is detected that is not compatible with the new version of Autodesk server, the existing database must be removed before the Autodesk server can be installed.

Operating System Requirements

Autodesk data management server supports both 32-bit and 64-bit operating systems. All operating systems listed in the following table are 32-bit. Supported 64-bit operating systems are also indicated.

Operating System (32-Bit)	Service Pack	64-bit	AWS	IIS
Windows® XP Home	SP2		•	
Windows® XP Professional	SP2	•	•	•

NOTE Limited to a maximum of 10 concurrent connections

Operating System (32-Bit)	Service Pack	64-bit	AWS	IIS
Windows® Vista™ Ultimate		•	•	•
Windows® Vista™ Enterprise		•	•	•
Windows® Vista™ Business		•	•	•
Windows® Vista™ Home Premium		•	•	
Windows® Vista™ Home Basic		•	•	

NOTE Limited to a maximum of 10 concurrent requests (data management queries). Windows Vista Home additions only allow 3 concurrent requests. Windows Vista Home additions should not be used as a data management server for more then one user.

Operating System (32-Bit)	Service Pack	64-bit	AWS	IIS
Windows server™ 2003 Server Standard	SP2	•		•
Windows server™ 2003 Server Standard R2	SP2	•		•
Windows server™ 2003 Enterprise	SP2	•		•
Windows server™ 2003 Enterprise R2	SP2	•		•
Windows server™ 2003 Small Business Edition	SP2			•
Windows server™ 2003 Small Business Edition R2	SP2			•

NOTE For Best performance on servers, we recommend a dedicated member (non-domain controller) server

The tables indicate the operating systems that support Autodesk Web Server (AWS) and Microsoft Internet Information Services (IIS).

For data management client add-ins, the system requirements are that of the design application.

Hardware Requirements

The following are minimum, recommended, and preferred requirements for running Autodesk data management server. If possible, meet the *preferred* requirements.

	Disk Space	Memory	Processor
Minimum	60 GB	1 GB	Intel® Pentium® 4, Intel® Xeon® AMD Athlon® 64 / Opteron, 2.0GHz

	Disk Space	Memory	Processor
Recommended	120 GB	2 GB	Intel Pentium 4, Intel Xeon AMD Athlon 64 / Opteron, 3.0GHz
Preferred	500 GB	4 GB	Intel Pentium 4, Intel Xeon AMD Athlon 64 / Opteron, 3.0GHz

Other Requirements

- Either a CD or DVD drive is required, depending on which media your software is delivered.
- Operating Systems installation media.
- Internet connection for Web downloads and Subscription Aware access.
- Microsoft Internet Explorer® 6 SP1 or later.
- We recommend settings that allow Microsoft Windows to manage virtual memory as needed. There should always be at least twice as much free hard disk space as system memory.

Installing Autodesk Data Management Solutions

2

If you are installing Autodesk® data management server for the first time, follow the procedures in this chapter to prepare for the installation. There are several factors to consider when determining the configuration of the system as well as important recommendations to help ensure a successful deployment.

NOTE If you are upgrading from a previous version of Autodesk data management server, skip this chapter and go to Upgrade to Autodesk Data Management Server 2009 on page 37. Important procedures are provided to ensure a successful upgrade.

Before You Start

Before installing the Autodesk data management server, you must ask yourself a few questions.

- How many users will need access to the vault?
- How many different geographical locations will need access to the vault?
- What other software will be loaded on the same computer as the data management computer?
- How you will manage server maintenance?
- How are your design teams and design projects are managed?

Knowing the answers to these questions will help choose the proper installation and system configuration for your data management server.

Quick Start

If you wish to jump directly to installation instructions refer to the table below to determine the best installation configuration for your situation.

Number of Users	Recommended Operating System	Recommended Web Server
One	Windows® XP	Autodesk® Web server (See Server installation for Windows XP)
One	Windows Vista™	Autodesk Web server (See Server installation for Windows Vista)
2-10	Windows server™ 2003	Internet Information Services (IIS) (See Server Installation for Windows Server 2003
10 +	Windows server 2003	Internet Information Services (IIS) See Server Installation for Windows Server 2003 **NOTE** Consider Upgrading SQL 2005 Express to Full SQL 2005

System Configuration

Autodesk recommends the server configuration described in this document. If you believe your system is better suited to a different configuration, we recommend that you consult with your Autodesk® reseller for a professional installation, and check the Autodesk Knowledge Base for any additional information regarding the upgrade.

Server Configurations

There are many ways to set up a vault server and its related clients. The most robust way is to install the Autodesk data management server on a separate

computer dedicated to the server and allow the appropriate clients to communicate with the server.

NOTE A single, dedicated server means that other competing business applications do not interfere with the operation of the Autodesk data management server. Other applications known to interfere are Microsoft® Exchange, using the computer as either an Internet or intranet server, and other data management and document management systems.

Maintenance Considerations

There are many other ways to configure and install Autodesk data management server. We recommend the single-server model for the following reasons:

Performance

Unless you have a very fast remote server and network, a single server hosting a local file store results in the best overall performance. If you have more then one geographical locations install the data management server at the main location first. If the performance is poor you can enhance the performance by using the multisite ability in Autodesk® Productstream®.

Reliability

More complex deployments are possible; however, the more complex the deployment, the more potential points of failure there are in your data management solution. The best data management deployment is a streamlined, efficient configuration on which you can rely for dependable data retrieval and maintenance.

Determining Vault Configurations

Before installing Autodesk® Vault, determine the installation scenario best suited for your company. Examples of scenarios are:

- A single design team

- Multiple design teams
- A single user
- Multiple remote sites sharing data

NOTE For more information on setting up a environment comprised of more than one site, see Setting Up Multiple Site Environments on page 75.

If you install a system that is too small, it can limit functionality or require reconfiguration in the future. There are several key factors to help you determine a scenario:

- Number of users in design team
- Number of design teams
- Availability of a network
- Availability of a server separate from user workstations
- Need to centrally administer vaults

After you choose the appropriate scenario for your situation, continue by installing and configuring the software components.

Working with Design Teams

When two or more users are actively sharing files on a design project, they constitute a design team. Designate a team member as the project leader. The team leader is responsible for the initial setup of the vault.

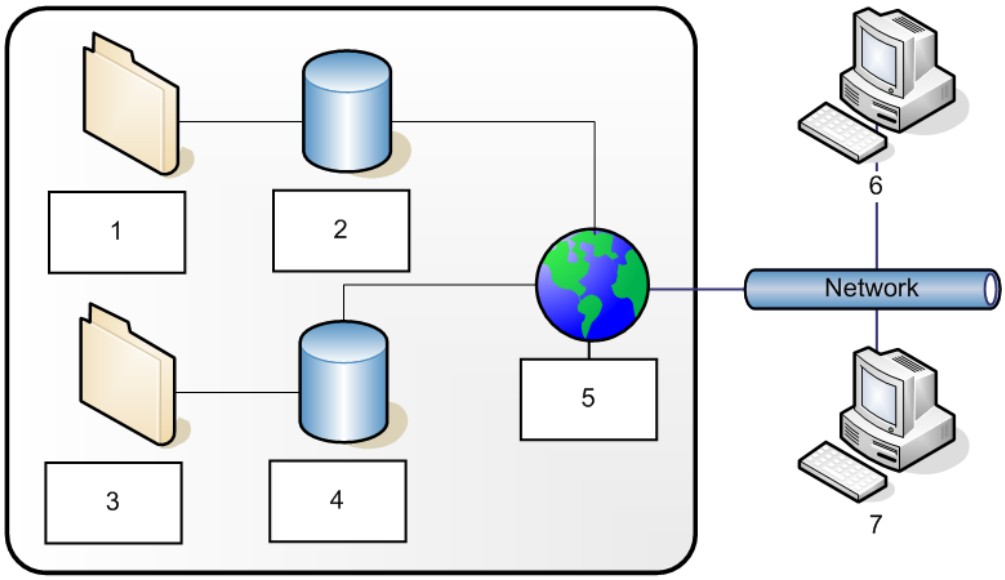
Key characteristics of a design team configuration:

- Network required (Windows file sharing capabilities, no domain required).
- A separate server is recommended for optimal performance, although one of the user computers can be used.
- Autodesk data management server components can reside on a Windows server operating system or on a Windows desktop operating system.

Key Characteristics of a multiple design team configuration:

- Two to three design teams that work independently of one another, and do not share files across teams.
- Network required (Windows file sharing capabilities, no domain required).
- A separate server is recommended for optimal performance. Installation of Autodesk® Inventor™ is not required on the server.
- Autodesk data management server components reside on a Windows server operating system.

To set up multiple design teams, use multiple instances of the scenario for a single design team, or install the Autodesk data management server components on a server separate from the user workstations. Each team member has the Autodesk Vault client components installed locally.



1. File Store "A"
2. Database "A"
3. File Store "B"
4. Database "B"
5. Web Server
6. Vault Client (Windows XP Pro Preferred)
7. Vault Client (Windows XP Pro Preferred)
8. Single Autodesk Data Management Server (Windows 2003 Server Preferred)

Working As a Single User

A single user working in a non-collaborative or isolated environment can install and run Autodesk data management server components and Autodesk Vault client components on the same workstation.

NOTE When working in a single user environment, it is not necessary to have Microsoft® Internet Information Services (IIS) installed. You can use the Autodesk Web Service included on the installation DVD/CD.

Key characteristics of a single user configuration:

- Single user (or multiple single users).
- Network not required.
- Separate server not required.

Vault Configurations

Depending upon the requirements of your business, you can configure your system with a single vault, or multiple vaults. The pros and cons of each are discussed next.

NOTE Using a single vault is the recommended method.

Single Vault Configuration

A vault is created automatically when Autodesk data management server is installed. Typically, use a single master vault for holding all of the design data for a design team, since data and folder mappings cannot be shared across vaults.

Multiple Vaults

Autodesk data management server can manage multiple vaults. A vault is a separate storage repository, similar to having multiple hard drives on a computer. The following section details benefits and downsides of using more than one vault.

Multiple Vault Advantages

You might choose to create additional vaults for the following purposes:

- To provide training areas where users can feel free to experiment without interfering with the main vault. It is possible to create a new vault for each user for training purposes, deleting the extra vaults when they are finished.

- To create additional vaults for test purposes, where users can test vault procedures without affecting the main vault.
- To create a separate vault for each customer, isolating data such as templates, standards, and project information for clients who each have separate libraries and shared content.
- To support different departments or individual teams working separate projects that do not need to share data.

Multiple Vault Disadvantages

The most important thing to remember when working with multiple vaults is that the data is completely separate and can never be merged. The only way to combine two or more vaults is to remove data from one vault and manually add it to the other, resolving any conflicts that may occur.

Multiple Remote Sites

Using Autodesk Productstream multiple remote sites can be configured to access the same vault data. This allows client applications located anywhere in the world to share the same vaults.

BEST PRACTICE Setup the master site and check the performance between the two sites via the network connection before setting up a multisite environment with replication.

NOTE For more information on installing Autodesk Productstream and setting up an environment comprised of more than one site, see Setting Up Multiple Site Environments on page 75.

SQL Server Upgrade

If you are currently using a previous version of MSDE in conjunction with Autodesk data management server, you should consider upgrading to either Microsoft® SQL Server 2005 Express or the full version of Microsoft SQL Server 2005. By upgrading from MSDE to a newer version of the SQL server, you eliminate the 10 user limit and increase the maximum database file size limitation of 2GB.

Consider upgrading to the full version of Microsoft SQL Server 2005 if you are using Microsoft SQL Server 2005 Express and the combined file size of the database files (*.mdf* and *.ldf*) for any single vault on the ADMS is approaching 4 GB.

NOTE It is important to monitor the size of the SQL database files. If you find your database approaching 2 GB using MSDE or 4 GB using Microsoft SQL Server 2005 Express, you should consider either removing unwanted properties from the system or plan to upgrade to the full Microsoft SQL Server 2005. Information about purging old file versions and removing unwanted properties is available in the ADMS Console Help.

There are two supported methods for upgrading or installing SQL Server for use with Autodesk data management server:

- Upgrade from Microsoft SQL Server 2005 Express to full SQL Server 2005.
- Install Autodesk data management server on an existing AUTODESKVAULT SQL instance.

Both methods can be found in the *Advanced Configuration Guide for Autodesk Data Management Server 2009*, accessible on the Autodesk Web site. Both methods are supported; however, the recommended method is to upgrade from SQL Server 2005 Express to full SQL Server.

Advanced Configurations

For more information about configuring Autodesk data management server, see the *Advanced Configuration Guide for Autodesk Data Management Server 2009* available in the online knowledge base at *www.autodesk.com*. The *Advanced Configuration Guide for Autodesk Data Management Server 2009* covers advanced topics such as:

- Upgrading from Autodesk Web Server to IIS
- Managing a remote file store
- Changing settings in configuration files
- Operating Vault on an existing SQL server instance
- Upgrading MSDE to full SQL server
- Running Vault with SSL

- Troubleshooting deployment issues

Server Installation

This section will give step by step instructions for setting up the data management server on Windows XP, Windows Vista, and Windows server 2003.

Windows XP

This section will explain the recommended steps to install the Autodesk data management server on a Windows XP workstation. The installation can be summarized in four major steps.

1. Choose the data management server hardware.
2. Load Internet Information Services (IIS)
3. Configure Windows XP SP2 Firewall
4. Run the data management server installation.

NOTE If 10 or more users will need access to the vault it is recommended to install the Autodesk data management server on Windows server 2003

What You will Need

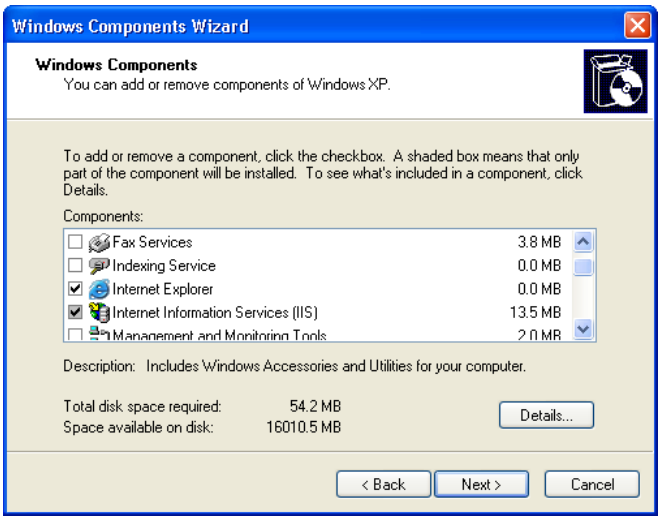
You will need the following items before you start.

- Autodesk data management installation media
- Windows XP installation media
- Administrator rights to the computer your installing the data management server on.

Installation

After choosing the data management server hardware. (see Hardware Requirements on page 11 for recommended specifications), you will need to load Internet Information Services on this system

1. From the Windows Start menu, select Programs ➤ Control Panel
2. From the Control Panel double click Add or Remove Programs
3. On the left side of the Add or Remove Programs dialog box select the option to Add/Remove Windows Components
4. Check Internet Information Services (IIS) and click the Next button. You may be prompted for your Windows XP CD.



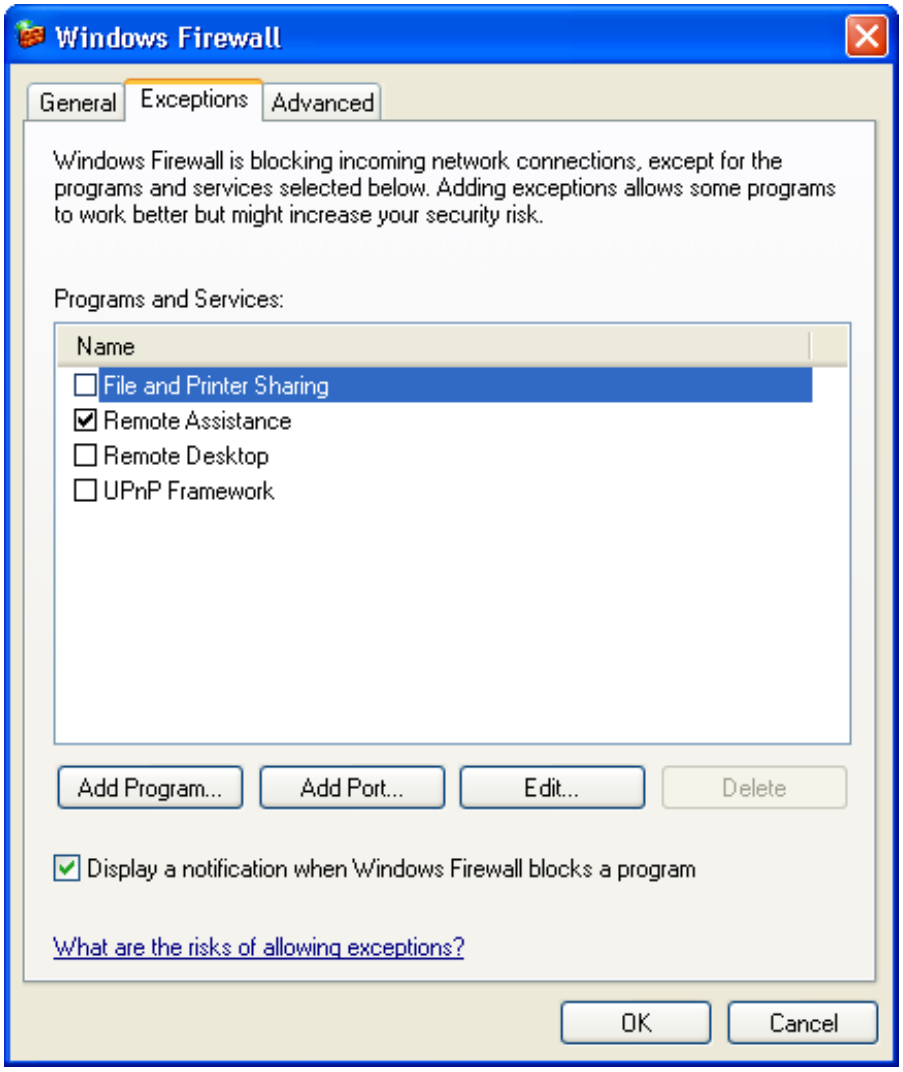
NOTE If IIS is installed and is not being used for any application uninstalling IIS and reinstalling it will ensure that the default settings are set

Now that IIS is installed you will need to configure the firewall that is built into Windows XP.

1. From the Windows Start menu, select Programs ➤ Control Panel
2. From the Control Panel double click Security Center

3 If the Firewall strip says on then click the firewall icon at the bottom of the dialog box.

4 In the Windows Firewall dialog box, click the Exceptions tab, and then click Add Port

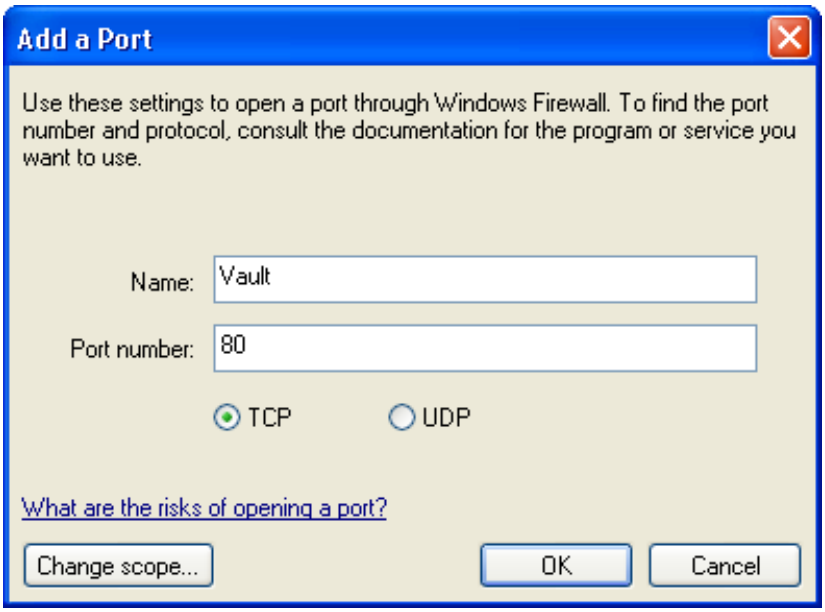


5 In the Add a Port dialog box, Enter the following information.

Name: Vault

Port: 80

Ensure TCP is checked



6 Click OK

NOTE If you are not running Windows XP Firewall you can skip this step.

Once the firewall is configured to allow Vault to communicate properly you will need to run the installation from the Autodesk data management server media.

1 Insert the Installation media and double-click the setup.exe

2 Choose Install Products.

3 Choose Autodesk Vault 2009 (Server).

4 Accept the License Agreement.

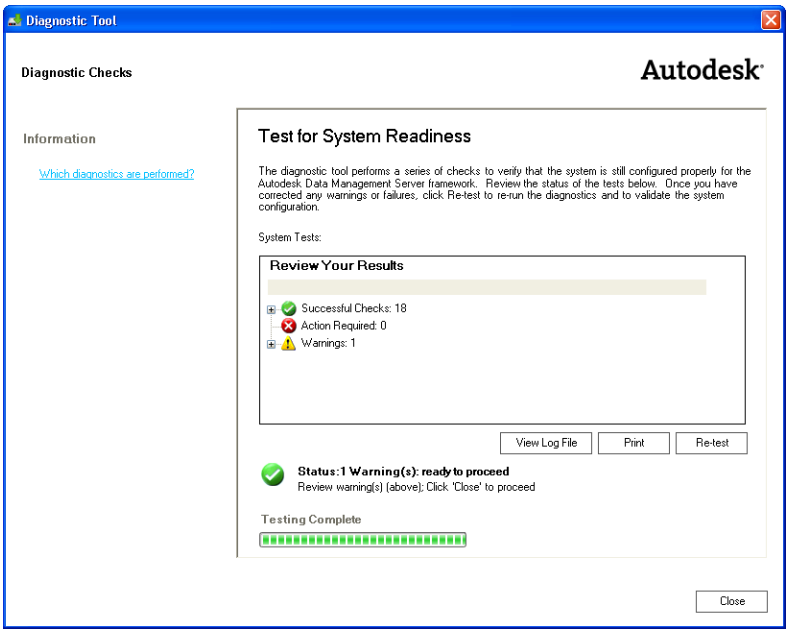
5 Fill in your user information.

6 The configuration screen allows you to change the location of the SQL installation, the SQL database location, and the location for the Autodesk data management server console application.

NOTE It is recommended to take the default locations unless you are setting up a remote site. For more information on Multi-Site setups see Setting Up Multiple Site Environments on page 75

7 Click Install

8 The installation will perform a series of installation pre-checks if your system does not pass the installation prechecks a dialog will appear displaying failures and warnings. Use the installation help and information in Troubleshooting on page 99



Windows Vista

This section will explain the recommended steps to install the Autodesk data management server on a Windows Vista workstation. The installation can be summarized in four major steps.

1. Choose the data management server hardware.
2. Load and configure Internet Information Services (IIS)
3. Configure Windows Vista Firewall
4. Run the data management server installation.

What You will Need

You will need the following items before you start.

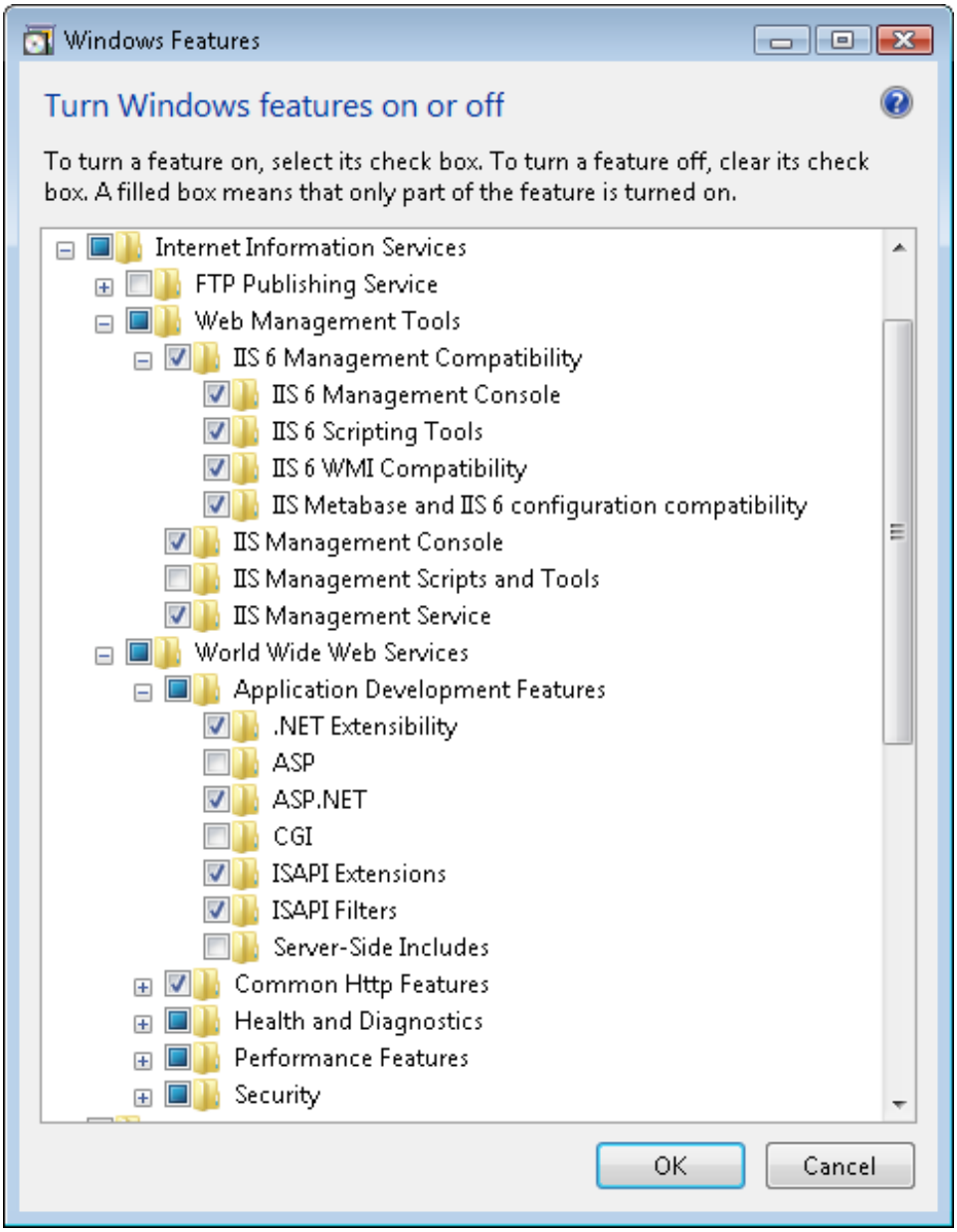
- Autodesk data management installation media
- Windows Vista installation media
- Administrator rights to the computer your installing the data management server on.

Installation

After choosing the data management server hardware. (see Hardware Requirements on page 11 for recommended specifications), you will need to load Internet Information Services on this system

1. From the Windows Start menu, select Programs ➤ Control Panel
2. From the Control Panel double click Programs and Features
3. On the left side of the Programs and Features dialog box select the option to Turn Windows Features on or off
4. Check Internet Information Services
5. Expand Internet Information Services ➤ Web Management Tools ➤ IIS 6 Management Compatibility.

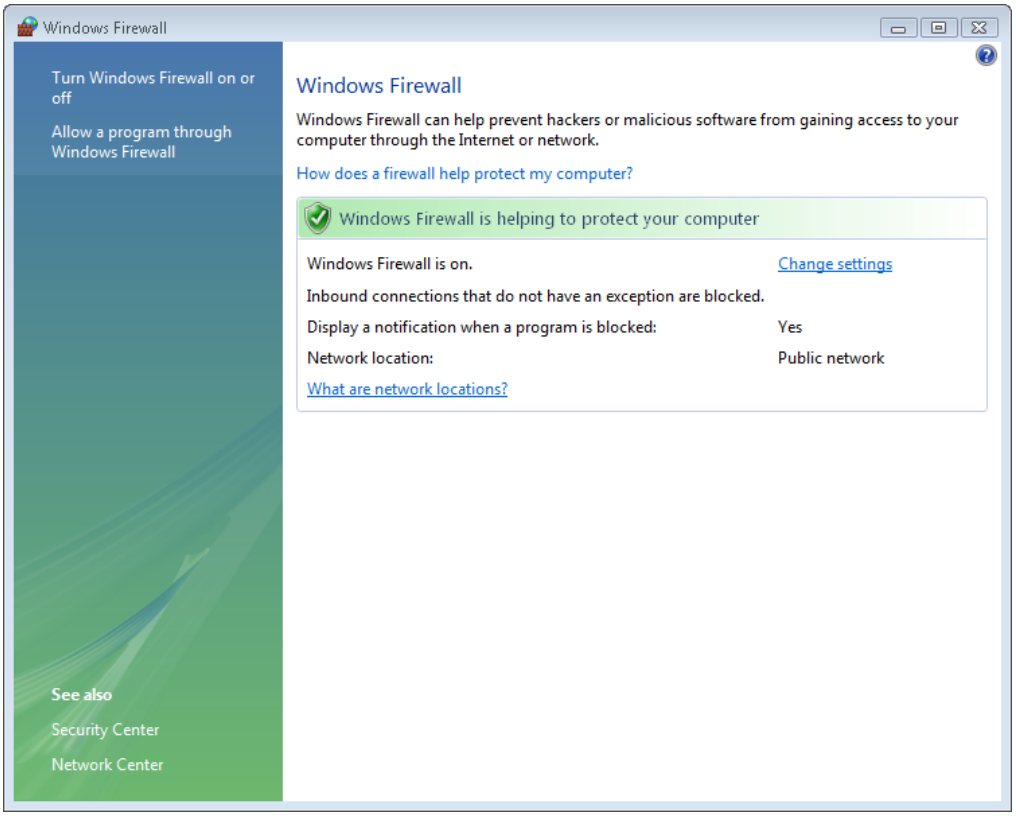
6. Turn on the check box for IIS Metabase and IIS 6 configuration compatibility.
7. Expand Internet Information Services ➤ World Wide Web Services ➤ Application Development Features.
8. Turn on the check box for ASP .NET. Three other settings are automatically turned on: .NET Extensibility, ISAPI Extensions, and ISAPI Filters


Windows Features
Turn Windows features on or off
To turn a feature on, select its check box. To turn a feature off, clear its check box. A filled box means that only part of the feature is turned on.
Internet Information Services
FTP Publishing Service
Web Management Tools
IIS 6 Management Compatibility
IIS 6 Management Console
IIS 6 Scripting Tools
IIS 6 WMI Compatibility
IIS Metabase and IIS 6 configuration compatibility
IIS Management Console
IIS Management Scripts and Tools
IIS Management Service
World Wide Web Services
Application Development Features
.NET Extensibility
ASP
ASP.NET
CGI
ISAPI Extensions
ISAPI Filters
Server-Side Includes
Common Http Features
Health and Diagnostics
Performance Features
Security
OK
Cancel

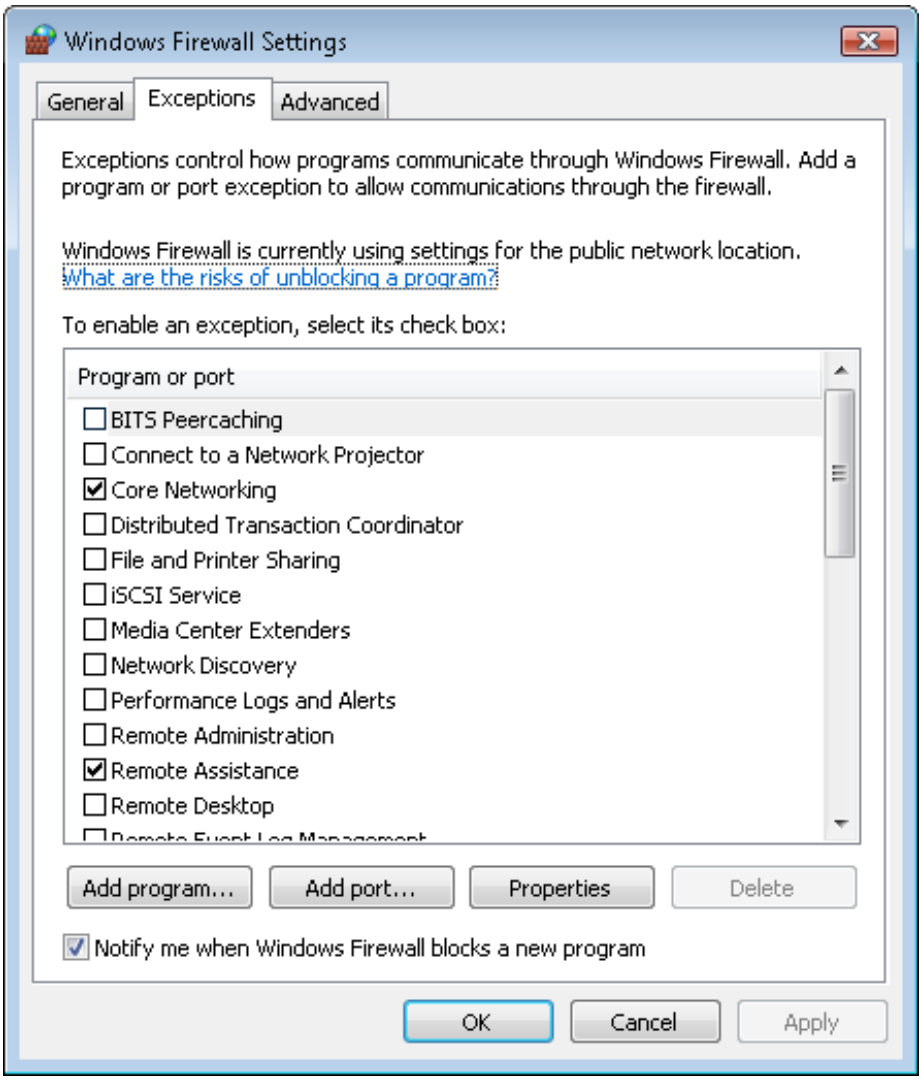
NOTE If IIS is installed and is not being used for any application uninstalling IIs and reinstalling it will ensure that the default settings are set

Now that IIS is installed you may need to configure the firewall that is built into Windows Vista

1. From the Windows Start menu, select Programs ➤ Control Panel
2. From the Control Panel double click Windows Firewall
3. If the Firewall strip says "Windows Firewall is helping to protect your computer," then click Change settings.



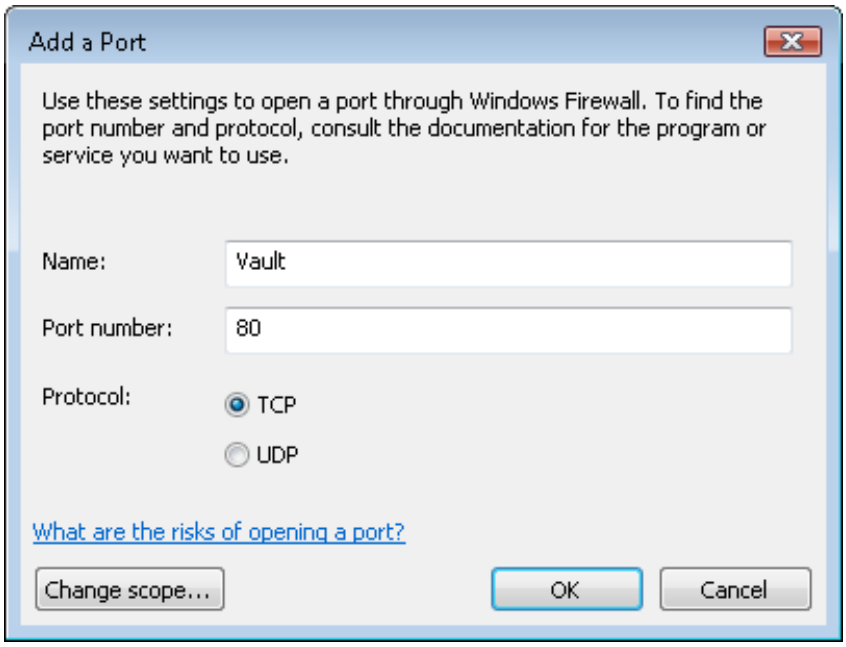
4. In the Windows Firewall dialog box, click the Exceptions tab, and then click Add Port



5 In the Add a Port dialog box, Enter the following information.

`Name: VaultPort: 80`

Ensure TCP is checked



6 Click OK

NOTE If you are not running Windows Vista Firewall you can skip this step.

Once the firewall is configured to allow Vault to communicate properly you will need to run the installation from the Autodesk data management server media.

1 Insert the Installation media and double-click the setup.exe

2 Choose Install Products.

3 Choose Autodesk Vault 2009 (Server).

4 Accept the License Agreement.

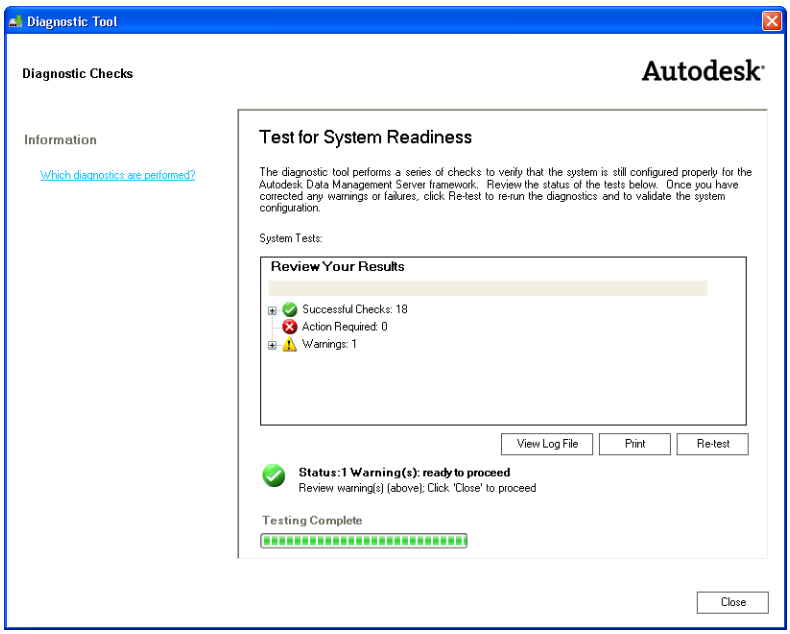
5 Fill in your user information.

6 The configuration screen allows you to change the location of the SQL installation, the SQL database location, and the location for the Autodesk data management server console application.

NOTE It is recommended to take the default locations unless you are setting up a remote site. For more information on Multi-Site setups see Setting Up Multiple Site Environments on page 75

7 Click Install

8 The installation will perform a series of installation pre-checks if your system does not pass the installation prechecks a dialog will appear displaying failures and warnings. Use the installation help and information in Troubleshooting on page 99



Windows Server 2003

This section will explain the recommended steps to install the Autodesk data management server on a Windows server 2003. The installation can be summarized in three major steps.

1 Choose the data management server hardware.

2 Load Internet Information Services (IIS) and ASP .NET

3 Run the data management server installation.

What You will Need

You will need the following items before you start.

- Autodesk data management installation media
- Windows 2003 Server installation media
- Administrator rights to the computer your installing the data management server on.

Installation

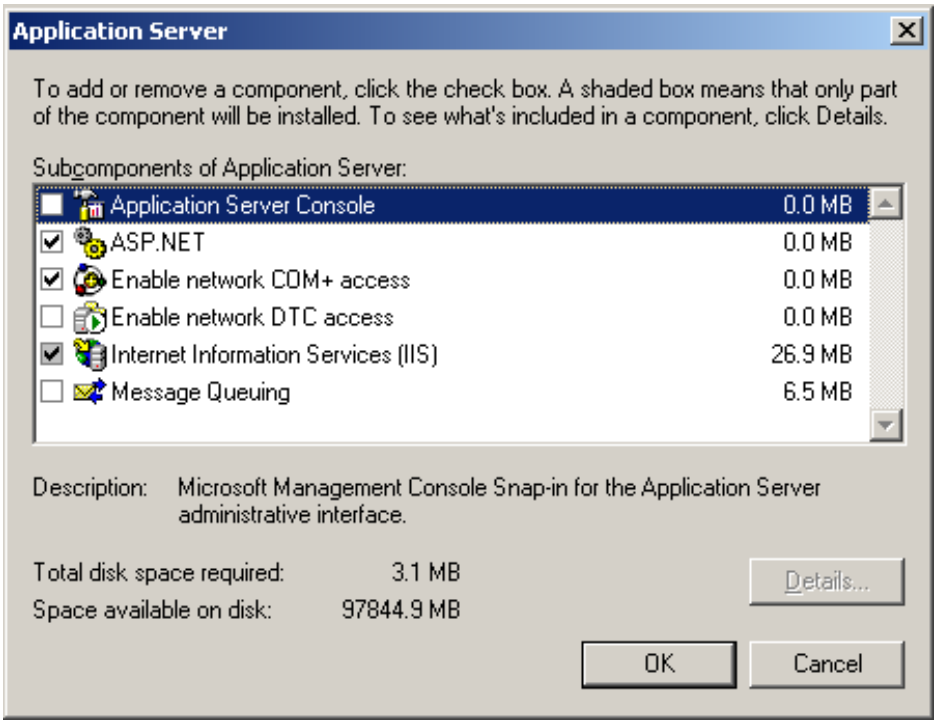
After choosing the data management server hardware. (see Hardware Requirements on page 11 for recommended specifications), you will need to load Internet Information Services on this system

1 From the Windows Start menu, select Programs ➤ Control Panel ➤ Add or Remove Programs

2 On the left side of the Add or Remove Programs dialog box select the option to Add/Remove Windows Components

3 Select and check Application Services and click the Details button

4 Check the ASP .NET option. This will cause the boxes for Enable network COM+ access and Internet Information Services to be checked as well.



5 Next button. You may be prompted for your Windows server 2003 CD.

NOTE If IIS is installed and is not being used for any application uninstalling IIs and reinstalling it will ensure that the default settings are set

---

Once the IIS is configured you will need to run the installation from the Autodesk data management server media.

1 Insert the Installation media and double-click the setup.exe

2 Choose Install Products.

3 Choose Autodesk Vault 2009 (Server).

4 Accept the License Agreement.

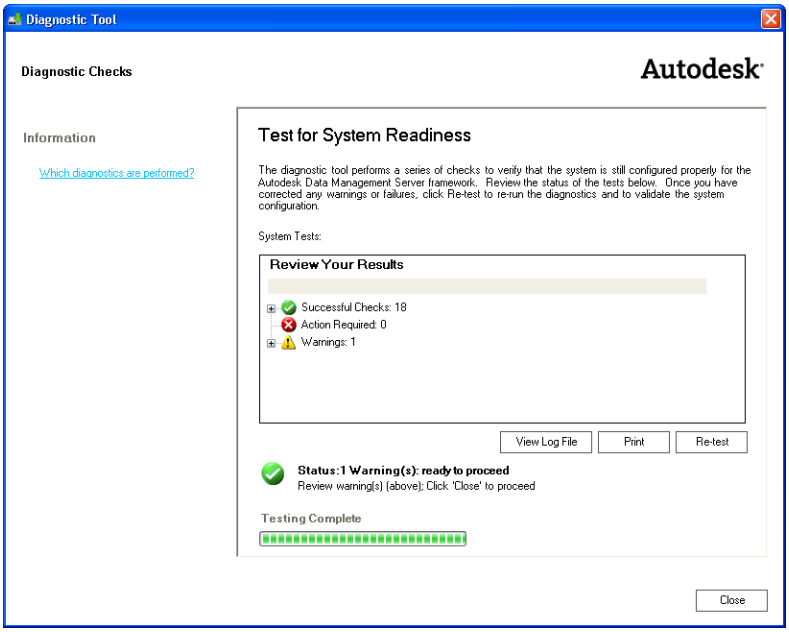
5 Fill in your user information.

6 The configuration screen allows you to change the location of the SQL installation, the SQL database location, and the location for the Autodesk data management server console application.

NOTE It is recommended to take the default locations unless you are setting up a remote site. For more information on Multi-Site setups see Setting Up Multiple Site Environments on page 75

7 Click Install

8 The installation will perform a series of installation pre-checks if your system does not pass the installation prechecks a dialog will appear displaying failures and warnings. Use the installation help and information in Troubleshooting on page 99



Upgrade to Autodesk Data Management Server 2009

3

One of the first steps to a successful deployment is to assess your organization's needs and ability to move to the next version. The checklists provide a starting point of some of the major areas that you should consider before moving to a new release.

Requirements

In this section, hardware and software requirements are listed in the form of checklists with which you can catalog your current system configuration. Additionally, a question and answer form is provided to help you assess your business requirements.

Hardware

The Autodesk® data management products have a server component and can support many client computers. Determine if the current computers meet the system requirements for the new release.

Understand the distinction between minimum, recommended, and preferred configurations. Minimum system requirements are suitable for testing or evaluation. Strongly consider using hardware that meets the recommended or preferred requirements for the planned system if you are supporting multiple users or using the system in production work. See Hardware Requirements on page 11.

Software

If you are planning to use the new version of Autodesk® Vault or Autodesk® Productstream® with a previous release of a CAD tool, then ensure that your client software is supported. In general, the data management products support the current CAD release and the previous CAD release. Check *www.autodesk.com* to make sure that your specific CAD tools are supported.

Business Requirements

One of the most important factors in determining how you want to approach upgrading your data management system is to understand the needs of your organization.

Question	Answer
Is the data in your vault critical to your organization?	
How long can you wait until the system is upgraded and available? Hours? Days?	
Are you rolling out any other software upgrades to your organization along with Autodesk Vault or Autodesk Productstream?	
Will you need to migrate any data that is already in a vault to a newer file format? (NOTE: This is different from migrating a vault itself)	

Testing

After assessing the hardware and software, determine whether you need a test environment. If you have any unanswered questions, or determine that the data management system is critical to your business, consider setting up a test

environment before upgrading to the next release of Autodesk Vault or Autodesk Productstream.

NOTE Using a second "test environment" is a best practice.

Test Environment

If possible, perform a test upgrade of the data management server on a machine that is similar to the one that is currently hosting the software. A test upgrade provides confidence that the backup is valid and helps you identify any potential pitfalls without impacting your production environment. It also helps you to determine compatibility with other applications and environment settings that your company uses.

Test Machine Settings	Benefits
Similar CPU, memory, disk space	Provides an estimate of how long the migration might take. Gives insight in to any migration settings such as time-out values that need to be changed. Shows other configuration settings that may require changes.
Database	Test machines that need to access a vault that is 2GB or more will also need to have a similar version of full Microsoft® SQL server installed.
Other applications that are installed on the production server	Test for compatibility with various proxy clients, firewalls, anti-virus programs, etc.
Operating System, Service Packs, firewalls, etc.	Compatibility testing.
Microsoft® Windows® Administration settings	Check for compatibility with Windows Domain policies, login scripts, permissions and other settings.

Migration

The Autodesk data management server console can migrate a vault from release 5 or 2008 to work with Autodesk data management server 2009. If you currently have a vault database earlier than release 5, migrate it first to release 5 and then use the server console to migrate it to work with Autodesk data management server 2009.

Before You Upgrade

When upgrading from a previous version of Autodesk data management server 2009, first back up your data and then uninstall and delete the components described in this section.

These are the steps required to completely remove Autodesk data management server 2008. Always create a backup of your vault data before uninstalling. Be careful to follow the steps outlined below in the order given.

Uninstall Components

Before uninstalling any components, back up existing vault and library data. Restore the backed up data on a test environment to verify that it can be successfully restored before proceeding.

NOTE Depending on your configuration, you may not find some of the components listed below. If they do not appear in the Windows Add or Remove Programs utility, skip the item.

- Autodesk data management server 2008 or earlier
- Autodesk Vault 2008 or earlier (if the server and clients are on the same computer)
- Autodesk Office Add-in (if the server and clients are on the same computer)

NOTE Do NOT uninstall the SQL instance AUTODESKVAULT.

Install Autodesk Data Management Server 2009

Once you have backed up the vault and library data, restored the backup on a test environment to verify it works, and removed previous versions of the Autodesk data management server and clients, you can install Autodesk data management server 2009. See Installing Autodesk Data Management Solutions on page 13 for instructions on installing the server.

During the installation, any existing databases and file stores are detected. You are prompted to migrate the data following the installation.

NOTE Vaults from Autodesk data management server versions 1.0 and 1.3 cannot be migrated. A vault from Autodesk data management server version 3 or 4 must be migrated to either version 5 or 2008 before being migrated to version 2009 or higher. An Autodesk Productstream vault cannot be migrated for use with Autodesk Vault.

When starting the server console the first time, it will ask you if you want to migrate any vaults it finds.If you Select no you will have to use the following procedure to migrate any existing vault.

Migrate existing vault data

1. From the Windows Start menu, select Programs ➤ Autodesk ➤ Autodesk Data Management ➤ ADMS Console.
2. The unmigrated vault is indicated with a yellow warning sign. Select the non-migrated vault.
3. Select Actions ➤ Migrate.
4. When the migration is done, click OK.

Installing Autodesk Data Management Clients

4

The Autodesk data management clients include the integrated add-ins for applications such as Autodesk® Inventor™ and AutoCAD®, as well as the stand-alone clients Autodesk® Vault Explorer and Autodesk® Productstream®. Add-in clients are available for Autodesk® and non Autodesk design applications. Once the Autodesk data management server is installed, the clients can be installed.

Data Management Client Requirements

Autodesk data management clients that integrate with an Autodesk design application require a licensed installation of the corresponding Autodesk application. Refer to the hardware and software requirements for each product.

The clients work in conjunction with the Autodesk data management server. The Autodesk data management server must be installed and configured before the clients can access the server.

NOTE The data management clients are not compatible with previous versions of Autodesk data management server. If you are upgrading to Autodesk Vault 2009 or Autodesk Productstream 2009, all server and client components must be upgraded.

Operating Systems

- Windows® XP Home (SP2)
- Windows® XP Professional (SP2)
- Windows Vista™ Ultimate (32-bit and 64-bit)

- Windows Vista™ Enterprise (32-bit and 64-bit)
- Windows Vista™ Business (32-bit and 64-bit)
- Windows Vista™ Home Premium (32-bit and 64-bit)
- Windows Vista™ Home Basic (32-bit and 64-bit)
- Windows XP Professional x64 Edition

Recommended Hardware

- Intel®Pentium® 4, Intel®Xeon®, ADM Athlon™ 1GHz
- 1Gb memory or higher
- 500Mb disk space

NOTE Although the Autodesk data management clients can be installed on the same computer with Autodesk data management server, it is recommended to install the server on a dedicated computer.

Before installing

- Close all Autodesk and Microsoft applications prior to installation.
- Disable any virus scanning software prior to installing.
- Previous versions of clients are not compatible with Autodesk data management server 2009. Installing Autodesk data management server 2009 requires that all clients are upgraded to version 2009.
- The recommended screen resolution is 1024x768 or higher. Large font settings in lower screen resolution settings may affect clarity of dialogs.

Client Installation

The Autodesk data management clients are available on the product installation media as well as on the Autodesk Vault 2009 media.

The Autodesk Vault add-in for Microsoft® Office was shipped with your software and can be used with one or more of the following Microsoft products as long as they are installed and registered on your machine.

- Microsoft® Word XP/2003/2007

- Excel® XP/2003/2007
- PowerPoint® XP/2003/2007

NOTE To install the Autodesk Vault clients, you must have either Microsoft Windows administrator privileges or power user privileges.

Installing the clients for an Autodesk design application

- Insert the Autodesk design application's installation media and follow the instructions for the installation wizard.

Installing the clients for a non-Autodesk design application

- Insert the application's installation media and follow the instructions for installing the software.

Installing clients from the Autodesk Vault 2009 Media

1. Insert the Autodesk Vault 2009 media.
2. Double-click *setup.exe*.
3. Follow the instructions in the installation wizard.

Add-in Integration

If a design application is already on the computer before the clients are installed, the corresponding add-in clients automatically integrate with the design applications when they are installed. However, if an application is installed after the data management clients, go to the Control Panel and modify the data management application to install the missing add-in.

First Log In

When Autodesk data management server is installed, a vault and a user account are created. The name of the vault is "Vault". The account name is "administrator". No password is assigned to the account and the name is not case sensitive. The administrator account has full administrator privileges.

NOTE If a user account has already been created for you, log in using your user name and password.

Log into Autodesk Data Management Server the first time

1 Use one of these methods:

- On the Start menu, click Programs ➤ Autodesk ➤ Autodesk Data Management ➤ Autodesk Vault.
- On the Start menu, click Programs ➤ Autodesk ➤ Autodesk Data Management ➤ Autodesk Productstream.
- In an application with an integrated add-in, click File ➤ Autodesk Data Management Server ➤ Log In.

2 In the Log In dialog box, verify the following:

User name	The name for the vault account.
Password	The password associated with the vault account.
Server	The name of the computer on which the vault server is installed.
Database	The name of a vault database located on the specified server. The default is "Vault". Click the browse button to select from a list of available databases on the server.

A default user ID, vault server, and database are set up for you to get started using the vault right away. If a user account has already been created for you, use your own account information. If a user account has not been created for you, or the default values are not present, contact your vault administrator.

Use either the default values or, if an account has been created for you, use your own account information. The default values are:

`User Name:` Administrator

`Password:` empty

`Server:` Name of the computer on which the vault server is installed. Use localhost if the server is installed on the same machine as the client.

`Database:` Vault

3 Click OK.

You are logged into the vault.

NOTE You have the option to save the account information to automatically log into the vault the next time.

Managing Autodesk Data Management Server

5

This chapter contains the procedures for Autodesk® data management server administration and management.

About Autodesk Data Management Console

A vault consists of three primary components:

- The master database called KnowledgeVaultMaster
- Vault database
- File store

KnowledgeVaultMaster is the master database for Autodesk data management server. Similar to SQL's master database, KnowledgeVaultMaster is a separate database containing common global administrative and configuration information for all of the vault and library databases on the Autodesk data management server.

The vault database stores information, such as file names, user names, properties, and file relationships.

The file store is a directory structure that contains the actual data files. Autodesk data management server console includes tools for maintaining databases and file stores.

NOTE Never directly move, delete, or edit a file in the file store.

All database and file store maintenance should be performed using the data management console. The console is the administrator's interface for maintaining the vault file stores and databases. Tasks include backing up and restoring data, creating new vaults, and purging old data. The administration utilities in the server console are also available from the command line. You can use the command line in conjunction with scripts to create custom schedules and automated maintenance tasks.

NOTE A database consists of an *.mdf* file and an *.ldf* file. These two files must be kept together for a vault to work correctly.

You must log into the server console with a vault account that has been assigned the Administrator role. In order to perform most vault maintenance, you must also have SQL administrator privileges. For more information about vault maintenance, command line options, and SQL administrator privileges, see the ADMS Console Help.

Start the Server Console

1. From the Windows® Start menu, select Programs ➤ Autodesk ➤ Autodesk Data Management ➤ Autodesk (Server Console).
2. In the Log In dialog box, enter the user name and password for the vault administrator account.
3. Click OK.

Performing Vault Maintenance

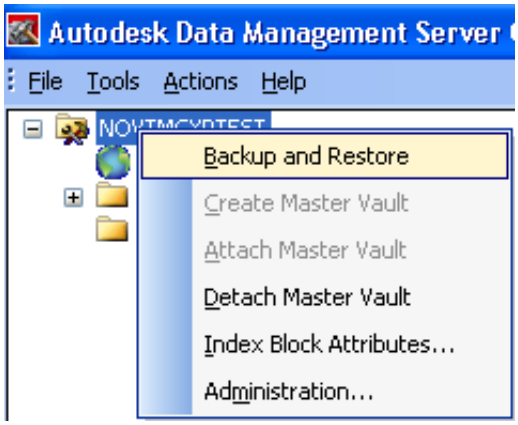
Ensure that no users are connected to the vault or library before performing any maintenance. Users are blocked from accessing the Autodesk data management server during maintenance procedures (Autodesk® Productstream® allows backups while users are connected).

Backing Up Vault Data

Backing up vault data is essential. The following are recommendations for backing up Autodesk® Vault.

Use the supplied backup-restore utility

Use the server console to backup all data necessary to restore a server if a failure occurs. When backing up and restoring the Autodesk data management server, no users can access any vault on the server unless the server is a Productstream server. When you back up the server, all users are blocked from accessing the system. In addition, server console backs up or restores all vaults on the server. There is no way to select individual vaults to back up or restore.



Develop a Backup Schedule

The next step is to automate the process. Two common methods are:

- Using the Windows Task Scheduler - this process uses the backup tools included with the server console as part of a standard Windows batch file.
- Including the backup as part of a tape backup set - this process uses a tape back up system to back up the vault directly or to use a method similar to the Windows Task Scheduler.

The preferred and most reliable method for backing up a vault is to integrate the server console backup tools into your tape backup plan.

With the server console command line options, you can create a script to automate the backup process. The following script is intended for a default Autodesk data management server installation. Edit the backup paths and installation paths as needed.

NOTE The following script is for performing full back ups ONLY. For more command line options or to create a script for performing incremental back ups, see the data management server console Help.

Automate backup using Microsoft® Windows® Task Scheduler

1. Create a new text file called *Backup.txt*.
2. Insert the following text:

   ```
   @ECHO OFF
   REM DELETE B AND CASCADE A BACKUP SUBDIRECTORIES
   RMDIR /Q /S "C:\Backup\Vault\B"
   REN "C:\Backup\Vault\A" “B”
   REM CREATE A NEW DIRECTORY FOR THE BACKUP
   MKDIR "C:\Backup\Vault\A\"
   REM START THE BACKUP PROCESS (THIS IS ONE LINE OF TEXT)
   "C:\Program Files\Autodesk\Data Management Server 2009\ADMS Console\Connectivity.ADMSConsole.exe" -Obackup -B"C:\Backup\Vault\A" -VUadministrator -VP[YourPassword] -S
   ```

3. Change the name of the text file to *Backup.bat* to convert it to a batch file.

NOTE When backing up a multi-site environment all backups should be done locally on the primary server.

Windows Vista User Account Control

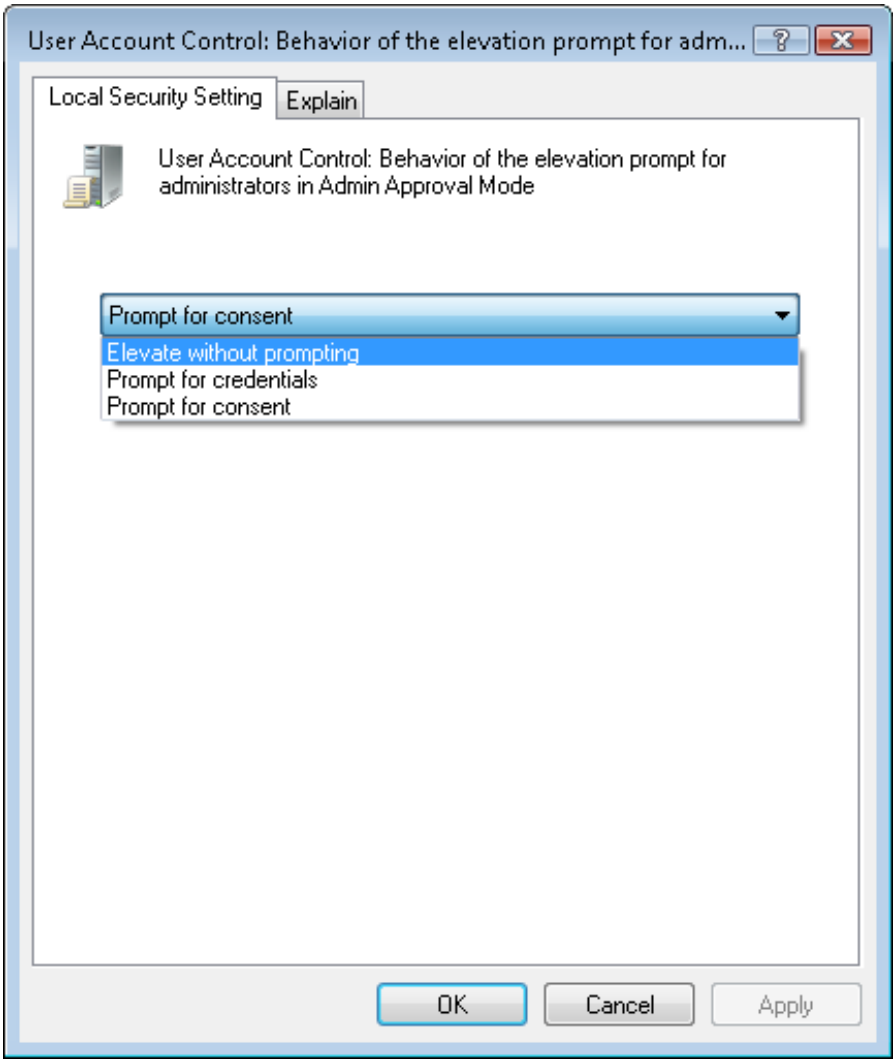
If you want to schedule automatic backups using the Autodesk server console, User Account Control settings in Windows Vista™ must be configured to allow applications to run without prompting for administrator permission.

NOTE The following settings affect the entire computer. All applications will run without prompting for administrator approval.

1. From the Windows Start menu select Run.
2. In the Run dialog box, enter secpol.msc and then click OK.
3. If you are prompted to allow the security policy application to run. Click Continue.
4. In the Local Security Policy dialog box, expand Security Settings ➤ Local Policies ➤ Security Options.

5 Locate the policy User Account Control: Behavior of the elevation prompt fro administrators in Admin Approval Mode. Right-click the policy and then select Properties.

6 In the Properties dialog box, select Elevate without prompting and then click OK.



7 Locate the policy User Account Control: Run all administrators in Admin Approval Mode. Right-click the policy and the select Properties.

8 In the Properties dialog box, select Disable and then click OK.

Creating Tape Backups

There are different ways to use a tape backup system to back up a vault. We recommend integrating the Server Console backup tools into your tape backup plan. Many systems allow you to run a script before and after the tape job executes. To back up a vault using a tape backup system, use the following scripts before and after the job runs.

NOTE We do not recommend using backup software to back up the SQL database and file store directly. This method increases the difficulty of migrating to a new release of Autodesk data management server or restoring a vault on a new machine. In addition, never back up the SQL Server using the "live" backup plug-ins that are available for major tape backup systems. You must ensure the file store and SQL data are perfectly in sync with each other. Failure to do so may render your data unusable.

---

1 Run the following script before your tape backup job starts:

```
@ECHO OFF

REM THIS WILL STOP THE WEB SERVER AND "CYCLE" THE SQL SERVER
REM PROVIDING THE BEST RESULTS FOR BACKING UP THE VAULT

IISRESET /STOP

NET STOP MSSQL$AUTODESKVAULT

NET START MSSQL$AUTODESKVAULT

REM START THE BACKUP PROCESS (THIS IS ONE LINE OF TEXT)

"C:\Program Files\Autodesk\Data Management Server 2009\ADMS
Console\Connectivity.ADMSConsole.exe" -Obackup -B"C:\Backup\Vault\A"
-VUadministrator -VP -S
```

2 Run the following script after your tape backup job completes:

```
@ECHO OFF

REM DELETE THE BACKUP SET AFTER THE TAPE SET COMPLETES

RMDIR /Q/S C:\Backup\Vault\A"

REM START THE WEB SERVER
```

IISRESET /START

3 Validate the Tape Backup

To ensure that your backup set will restore properly, test both the backup and the backup procedures. We recommend that you install the Autodesk data management server on a separate computer and test a full restore of the data using the Restore command in Autodesk data managment server console.

Restoring Vaults from Back Ups

1 Click Tools ➤ Restore.

2 Restoring a vault deletes the current data sets and file store. This action cannot be undone. You are prompted for confirmation before proceeding. Click Yes.

3 Select whether you are restoring a directory or a file.

4 In the Restore from directory field, specify the location of the backed up data. To browse for a location, click ... and locate a directory using the file browser.

5 Select whether the database is to be restored to the original location or to a different location. If you choose Select Restore Location, specify a target directory for the database. This may be necessary when restoring data to a different machine that doesn't have the same drive letters or locations available.

6 Select whether the file store is to be restored to the original location or to a different location. If you choose Select Restore Location, specify a target directory for the file store.

7 Click OK.

The vault data is automatically migrated when it is restored using Autodesk data management server console. If you are restoring the data using the command line, you must migrate the data after it is restored.

NOTE Back up files created from the Autodesk data management server installed with using Autodesk Productstream cannot be restored to an Autodesk data management server installed with Autodesk Vault.

Managing User Accounts

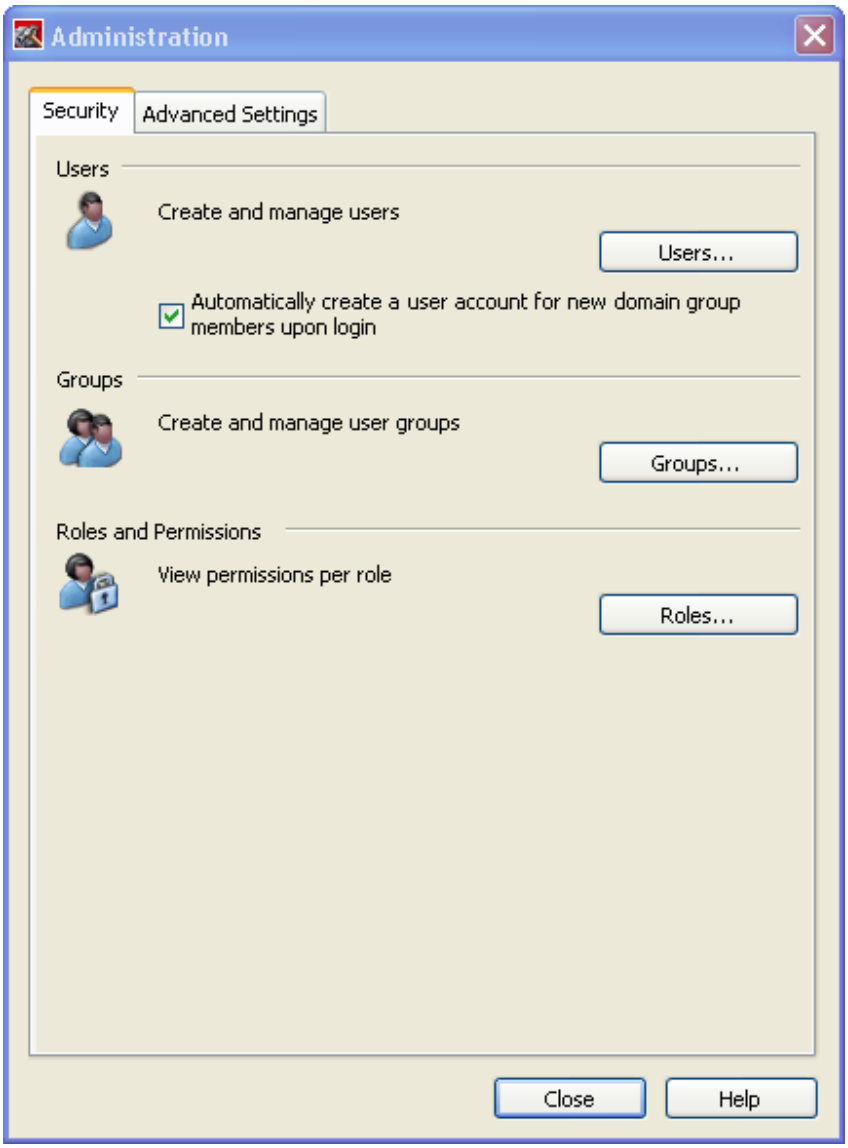
The vault is a secure database. Before anyone can use a vault, the administrator must add accounts for users and grant levels of permission within the vault using roles.

For more information about users, roles, and managing accounts, see the server console Help.

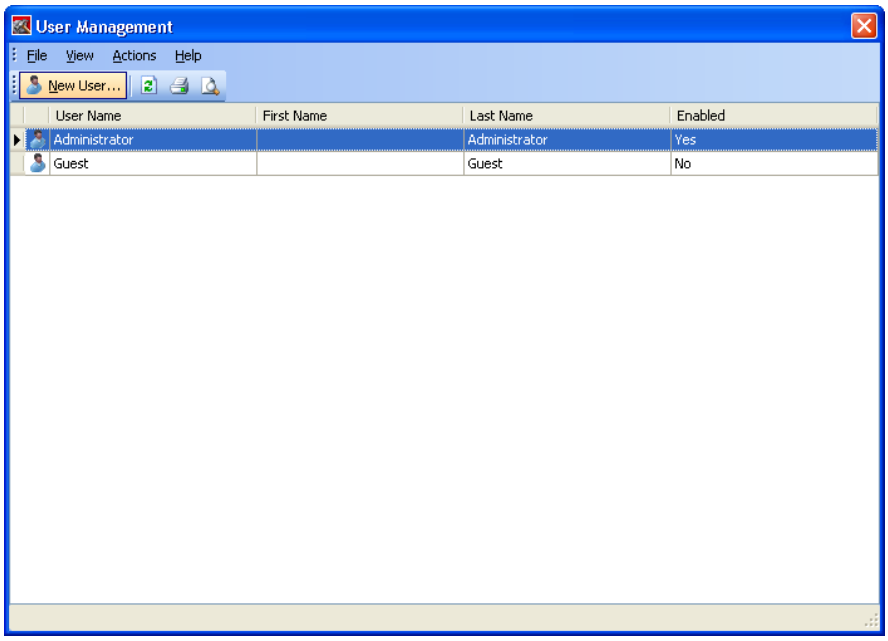
Adding or Editing User Accounts

You can create user accounts and grant vault access.

1. Click Tools ➤ Administration.
2. In the Administration dialog box, click Users.

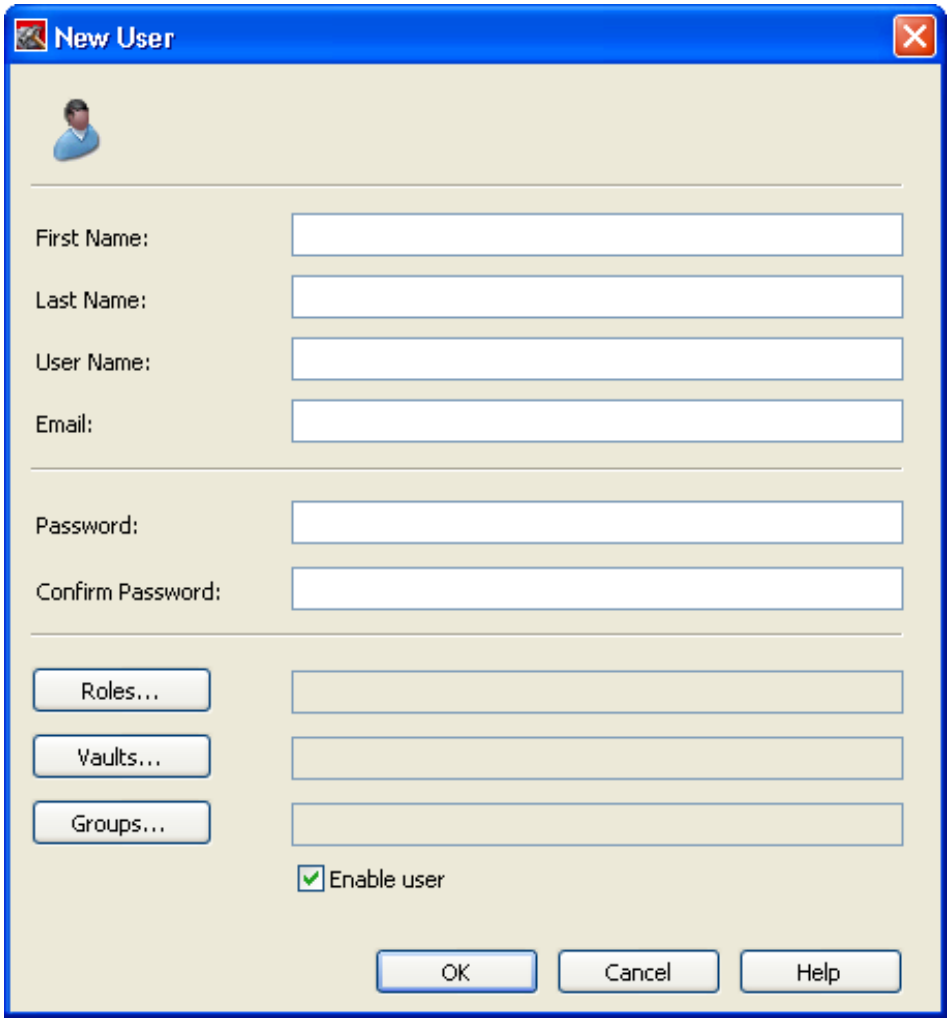


3 In the User Management dialog box, click New User.



4 In the New User dialog box, enter the information for the new user:

- First name
- Last name
- User name
- Email address



5 Enter a password and confirm it.

6 Click Roles and assign one or more roles to the account. Roles determine the level of access to the vault.

7 Click Vaults and select one or more vaults for the account to access.

8 Select the Enable User check box. Until the account is enabled, it is not available for use and cannot access the vault

Repeat this process for each new user account needed. Each new user is granted access to the selected vaults on the server.

Edit a user account

1. To edit an existing user's account, double-click a user's profile in the User Management dialog box. Or highlight a user's profile, and then click Edit User.
2. In the User Profile dialog box, edit the information for the current user.
3. To edit the password, enter a new password and then confirm the new password.
4. Click Roles to edit the user's roles.
5. Click Vaults to edit the user's access to vaults.
6. Click OK.

NOTE Each user logs into the vault with a unique user name and password. Restrict access to the administrator account to only the assigned vault administrator.

Assigning Roles to User Accounts

1. Select Tools ➤ Administration.
2. In the Administration dialog box, click Users.
3. To assign roles to a user for the first time, click New User.
4. In the New User dialog box, select "Roles.
5. In the Add Roles dialog box, select one or more roles, and then click OK.
6. To edit an existing user's role, double-click a user's profile in the User Management dialog box.
7. In the User Profile dialog box, click Roles.
8. In the Add Roles dialog box, select or cancel the selection of one or more roles.
9. Select the Enabled check box. Until the account is enabled, it is not available for use and cannot access the vault.
10. Click OK.

More than one role can be assigned to a user account. The permissions are a union of the assigned roles.

NOTE The Content Center Administrator, Content Center Editor, and Content Center Reviewer roles are specific to Content Center and should not be altered.

Granting Vault Access to Users

1. Select Tools ➤ Administration.
2. In the Administration dialog box, click Users.
3. To grant a user access to one or more vaults for the first time, click New User.
4. In the New User dialog box, click Vaults.
5. In the Add Vaults dialog box, select one or more vaults, and then click OK.
6. To edit access to a vault, double-click a user's profile in the User Management dialog box.
7. In the User Profile dialog box, click Vaults.
8. In the Add Vaults dialog box, select or cancel the selection of one or more vaults.

Disabling User Accounts

1. Select Tools ➤ Administration.
2. In the Administration dialog box, click Users.
3. In the User Management dialog box, double-click a user's profile you want to disable, or highlight the user account and click Edit User.
4. In the User Profile dialog box, clear the check box of the Enable User.
5. Click OK.

Managing Vaults

A vault is created automatically when Autodesk data management server is installed. Typically, use a single vault for holding all of the design data for a design team, since data and folder mappings cannot be shared across vaults. However, you may want to create additional vaults for:

- testing purposes so users can become familiar with vault procedures without affecting the main vault
- contractors who want to use multiple vaults to isolate data from completely independent clients who each have separate libraries and shared content

Create a vault

1. In Autodesk data management server console, click Vaults.
2. Click Actions ➤ Create Vault.
3. Enter a name for the new vault. The name cannot contain certain special characters.
4. The file store can be located in a default location, or a new location can be specified. Click OK to create the new vault with the default file store location or click Select File Store Location to specify a new location and then click OK.

NOTE The default path for the file store is *C:\Documents and Settings\All Users\Application Data\Autodesk\VaultServer\FileStore*. The default path for the file store on a Windows Visa™ system is *C:\Program Data\Autodesk\VaultServer\FileStore.*

Deleting Vaults

Vaults that are no longer in use can be deleted. Before deleting any vault, back up the data.

1. In Autodesk data management server console, click Vaults.
2. Select a vault to delete.
3. Click Actions ➤ Delete Vault.
4. You are prompted to verify your actions. Click OK.

Moving File Stores

While it is recommended to keep all of the Autodesk data management server components on a single machine, the file store can be moved from its existing location to another location. To move a file store to a location on a remote machine, see "Managing a Remote File Store" in the Advanced Configuration Guide available in the online Vault Knowledge Base.

NOTE Use only the data management server console to move a file store.

Attaching and Detaching Vaults

Detaching a vault disconnects it from the server so it can be moved to another location. The detached vault should not be moved off of the machine hosting the SQL server. Before detaching the master vault database, you must detach all other databases. The vault and its log file are detached. The names of the files for a vault named Vault are:

- *Vault.mdf*
- *Vault_log.ldf*

After the vault is detached, you can move the two corresponding files. The *.mdf* and *.ldf* files must be kept together. Attaching a vault reconnects the set of vault files that were moved to the database engine.

NOTE After re-attaching the database, the Autodesk data management server user accounts must be edited to have access to the re-attached vault. To avoid editing the user accounts, back up the database and restore it to a new location.

Detach a vault

1. Ensure that no users are accessing the vault (unless using Productstream).
2. In Autodesk data management server console, select a vault from the main view.
3. Click Actions ➤ Detach.
4. You are prompted to confirm your actions. Click Yes.

Detach the master vault

1. Ensure that no users are accessing the vault (unless using Productstream).
2. Click Tools ➤ Detach Master Vault.
3. You are prompted to confirm your actions. Click Yes.

NOTE You must detach all other databases before detaching the master vault.

Attach a vault

1. Select the Vaults folder.
2. Click Actions ➤ Attach.
3. In the Attach Vault dialog box, enter the name of the data file (*.mdf*) or click the browse button to locate the file.
4. Enter the name of the log file (*.ldf*) or click the browse button to locate the transaction log file.
5. Enter the name of the file store or click the browse button to locate the file store.
6. Enter the name of the vault.
7. Click OK.

For more information on detaching and attaching vaults, see the Autodesk Vault Help or visit the Autodesk Web site at *www.autodesk.com*, and search for support document TS81609.

Purging Vault Data

To help maintain drive space and to minimize file store size for better performance, you can purge old file versions and out-of-date information from the vault. You specify the conditions that must be met in order for a file version to be kept in the vault. Versions that do not meet the specified criteria are then removed from the vault.

Basic rules of purging

- The latest version cannot be purged. To remove a file and all of its versions from a vault, use Delete.

- File versions linked to items in Autodesk Productstream cannot be purged. To remove a version linked to an item, the item must first be deleted from the item master.
- File versions that are labeled cannot be purged.
- Locked files cannot be purged.
- If a file to be purged is checked out, an extra version is maintained.
- Children with a dependent parent version cannot be purged until the parent version has been purged.

1. Select a vault from the main view.
2. Click Actions ➤ Purge.
3. In the Vault Version Purge dialog box, turn on the check box next to each of the selection rules to determine the data to be purged:

Versions except latest	Specify how many file versions to leave in the vault.
Versions older than days	Specify how old the files to purge should be.
Exclude versions where comment contains	Enter a text string contained within the Comments field of the files in the vault. Any files containing this string will be excluded from the purge process.

4. Click OK.

Re-Indexing the Server

Re-Indexing the server extracts the properties from all the files in the vault. Typically, this indexing process happens when a file is first added to the vault or when an updated version is checked in. From time to time it may be necessary to re-index the entire vault to:

- Add new properties to the vault that were previously unknown to the system.
- Remove unwanted properties from the vault that are no longer needed, in turn saving space in the SQL Sever database.

The Autodesk data management server utilizes a technology created by Microsoft called iFilters. iFilters are small applications that exist on the server

containing a set of instructions on how to open and extract property data from files. A few iFilters are installed with Autodesk data management server by default and other exist with the operating system. There are also many third party iFilters available for other file formats. An Internet search for iFilter returns some of the filters that other software developers have published.

The iFilters that are included by default index the following file types:

- AutoCAD®*.dwg*
- Autodesk®Inventor™
- Autodesk® 3D Studio Max®
- Microsoft® Office

Adding New Properties

When a new iFilter is installed, the vault database must be re-indexed to recognize the related properties. After, run the Re-Index command to extract the new property data from all of the files in the vault, including the historical versions.

NOTE Some iFilters create actual properties that are searchable and can be displayed as columns in the vault. Others only index file contents which are searchable, but cannot be displayed as a column. For more information on searching file contents, see the next section Full Content Indexing on page 67.

Removing Unwanted Properties

Autodesk data management server automatically extracts all the possible properties it can from the system, making them available to users. It may become necessary over time to remove some of these properties from the system to either prevent users from accessing them (reducing clutter) or to actually reduce the amount of space that SQL Server uses to store these properties.

Removing properties from the system requires first using Autodesk Vault Explorer or Autodesk Productstream's File Properties management inside of the administration tools.

After configuring the properties display via the In Use/Not in Use settings, you are able to re-index the vault, removing any property from the system that was marked Not In Use.

Property maintenance should be performed occasionally. If for any reason you decide at a later time that a previously removed property is needed again, you can mark the property In Use and re-index the database.

For more information on re-indexing properties and property maintenance, see the server console Help.

Full Content Indexing

Full content indexing is the ability to index the actual file contents of many different document types. For example, reading simple text on a *.dwg* file or finding all occurrences of a word in Microsoft® Word documents.

Enabling Full Content Indexing

By default, full content indexing is not enabled. Full content indexing can be enabled per vault on the Autodesk data management server.

1. In the server console, select a vault from the navigation pane.
2. Select Actions ➤ Content Indexing Service.
3. In the Content Indexing Service dialog box, select Yes, enable Content Indexing Service.

Microsoft Indexing Service

Unlike the standard property indexing performed by the Autodesk data management server, full content indexing is done using the Windows Indexing Service. The Windows Indexing Service is enabled by default.

If the Windows Indexing Service needs to be installed

1. From the Windows Control Panel, double-click Add or Remove Programs.
2. In the Add or Remove Programs dialog, click Add/Remove Windows Components.

3 In the Windows Components Wizard, turn on the Indexing Services check box and then click Next.

4 When the component is installed, click Finish.

When you enable the full content indexing for a vault, a new catalog is created in the Index Server prefixed with ADSK. This is the repository for indexed contents.

The Microsoft Indexing Service is implemented as a shared server. There is no degraded performance with check-in times, nor will the database size increase. However, a full content search will take slightly longer to return results when compared to a more simple property based search.

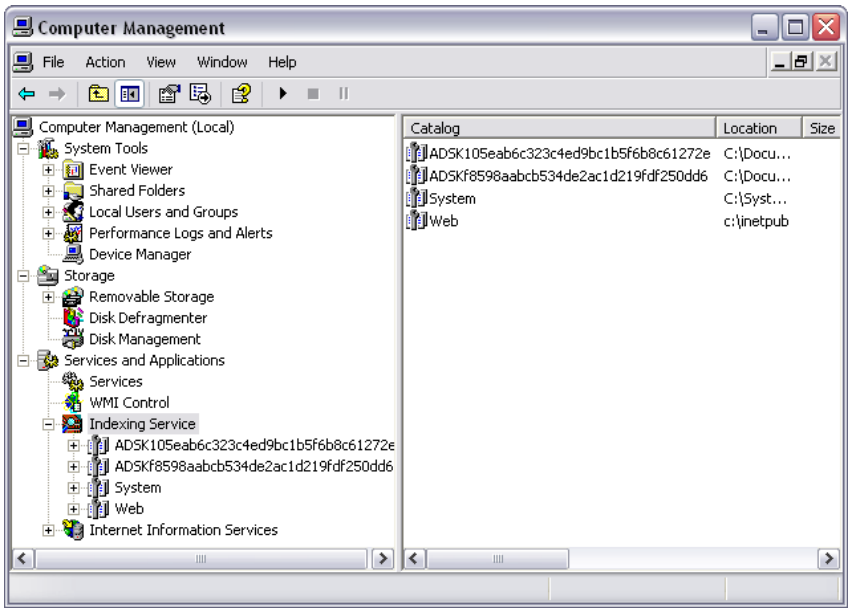
NOTE The server console manages the Indexing Service under normal operation.

Access the Windows Indexing Service

1 From the Windows Control Panel, double-click Administrative Tools.

2 In the Administrative Tools window, double-click Computer Management.

3 In the Computer Management dialog box, expand Services and Applications ➤ Indexing Service.



Managing Content Center Libraries

In addition to hosting vaults, Autodesk data management server also hosts Content Center libraries. Use the server console to perform Content Center Library administrator tasks.

For information on installing Content Center libraries, refer to the Content Center Installation Guide available on the Autodesk Inventor 2009 installation media, disk 1.

Maintaining Libraries in ADMS Console

Use the server console to create, manage, and delete libraries. The console displays the list of Content Center libraries currently installed on the server.Using the Backup command, you can create a back-up of an existing library on the server. Select a library from the list to manage it.

A Content Center library is contained within a SQL Server database. The database is comprised of a pair of files with the extensions MDF and LDF. These files together make one library and need to be maintained together.

Creating New Libraries

You can create a new read-write or read-only Content Center library, which can be configured for use in either a local or an Autodesk Inventor project that accesses the Content Center libraries from a central server.

Create a library

1. In the server console, select the Libraries folder in the navigation pane.
2. Click Actions ➤ Create Library.
3. Enter a name for the new library.
4. Select a library partition from the drop-down list.
5. Enter a name for the library database.
6. Click OK.

Deleting Libraries

Deleting a library detaches the selected library from the database server, and deletes the data files (*.mdf* and *.ldf*) from the hard disk. All data contained in the library is permanently deleted.

Delete a library

1. In the server console, select a library you want to delete in the navigation pane.
2. Click Actions ➤ Delete Library.
3. Click OK.

Attaching Libraries

Use Attach to make a Content Center library available on the server from a pair of existing *.mdf* and *.ldf* data files already on the computer.

Attaching a library

1. In the server console, select Libraries in the navigation pane.
2. Click Actions ➤ Attach Library.
3. Click the browse button, and locate the *.mdf* file you want to attach. The *.ldf* file will automatically be attached.
4. In the Attach Library dialog box,
5. Click OK.

Detaching Libraries

Detaching a library removes a Content Center library from the list of available libraries in the server console. Detach does not delete its data files (MDF and LDF) from the server. The data files can be re-attached and made available at a later time using the Attach command, or can be manually copied to another location, for example, to be backed up, or moved to another machine.

Detaching a library

1. In the server console, select Libraries in the navigation pane.
2. Click Actions ➤ Detach Library.
3. Click the browse button, and locate the *.mdf* file you want to detach.
4. Click OK.

Exporting Libraries

Export copies a set of library files (*.mdf* and *.ldf*) that are in the server console to a specified location. You can specify a folder on the same machine, or a location available on the network.

Export a library

1. In the server console navigation pane, select a library you want to export.
2. Click Actions ➤ Export Library.
3. In the Browse for Folder dialog box, specify the location you want to export the *.mdf* file to, and click OK.

Importing Libraries

Import copies library files (a pair of MDF and LDF files) into the standard SQL Server data files directory, and then performs an attach operation to register them with SQL Server and make them available as a Content Center Library. For example, use Import to copy library files from a 3rd party supplier to the Content Center Library you are administrating

Import a library

1. In the server console navigation pane, select a library you want to import.
2. Click Actions ➤ Import Library.
3. In the Browse for Folder dialog box, locate the *.mdf* file you want to export, and click OK.

Library Summary

Highlight a library to display high-level information about the library and its data files in the main pane.

Library Status

Right-click on a library in the browser and select Read-Only to toggle the status of a library to be marked as Read-Only or Read-Write. You cannot change the status of a default read-only libraries installed with Autodesk Inventor.

Content Center Permissions

If you require editing permissions for one or more Content Center libraries, you must have a Content Center Editor account set up for you in the data management server console.

Accounts are not required for users who only need read-only access to the Content Center library.

To create Content Center library user accounts in the data management server console

1. In server console, click Tools ➤ Administration.
2. In the Administration dialog box, click Users.
3. In the User Administration dialog box, click New users.
4. In the New User dialog box, enter the desired information for the user. Click the Roles button to assign the user a Content Center role.
5. Select Content Center Editor to assign editing permissions.
6. Click Vaults to assign the user to a vault. You must assign a user to a vault even if you are not using Autodesk Vault Explorer. This is required in order to log into the Autodesk data management server in Autodesk Inventor to access the Content Center libraries.
7. If needed, click Groups to assign the user to a group (optional).
8. Click OK.

9 In the User Management dialog box, and then in the Administration dialog box, click Close.

10 Communicate user account information and the Login procedure to each member in the team.

Learning and Training

Before getting started and rolling out an Autodesk data management solution to all the users in your group, we advise that you learn the basics and understand your deployment approach. Since Autodesk Vault can affect multiple users, you will want to make sure that the deployment goes as smoothly as possible.

Learning to Use Autodesk Vault

NOTE It is recommended that all users go through training from an Autodesk Authorized Training Center in addition to the steps below.

As a first step to learning to use Autodesk Vault, refer to the tutorials that are included in with the Autodesk Vault Help. There is a tutorial for each of the design application add-ins. These tutorials are accessible from the respective Help menus in the design application.

Search the Autodesk Knowledge Base:

- Knowledge Base *www.autodesk.com/vault-support*

More online information about Autodesk Data Management Server and client applications:

- Autodesk Inc. *www.autodesk.com*
- Autodesk Vault *www.autodesk.com/vault*
- Autodesk Productstream *www.autodesk.com/productstream*
- Autodesk Discussion Group *www.autodesk.com/discussiongroup-vault*
- Autodesk Discussion Group *www.autodesk.com/vault-whitepapers*

NOTE Autodesk sponsored discussion groups are a forum where Autodesk customers and industry partners can ask questions and share information about Autodesk products. Discussion group topics will be publicly available worldwide. You are welcome to participate by posting questions or providing input or answers to questions from other users.

Getting Training

Be more productive with Autodesk software. Get trained at an Autodesk Authorized Training Center (ATC) with hands-on, instructor-led classes to help you get the most from your Autodesk products. Enhance your productivity with proven training from over 1,400 ATC sites in more than 75 countries. For more information about Autodesk Authorized Training Centers, contact *atc.program@autodesk.com* or visit the online ATC locator at *www.autodesk.com/atc*.

Training Other Users

As previously mentioned, when training other users on Autodesk Vault, we advise that you use a new, personalized vault for learning purposes. Since a new vault is 100% independent of others, there is no risk that their learning process will interfere with production data.

Setting Up Multiple Site Environments

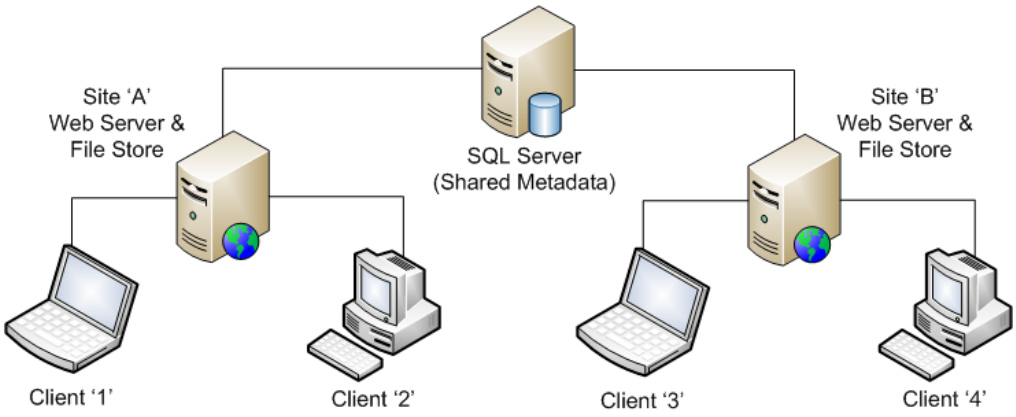
6

Multiple remote sites can be configured to access the same vault data. A site is comprised of a dedicated Autodesk® data management server and a file store. Each site accesses a shared AUTODESKVAULT SQL instance.

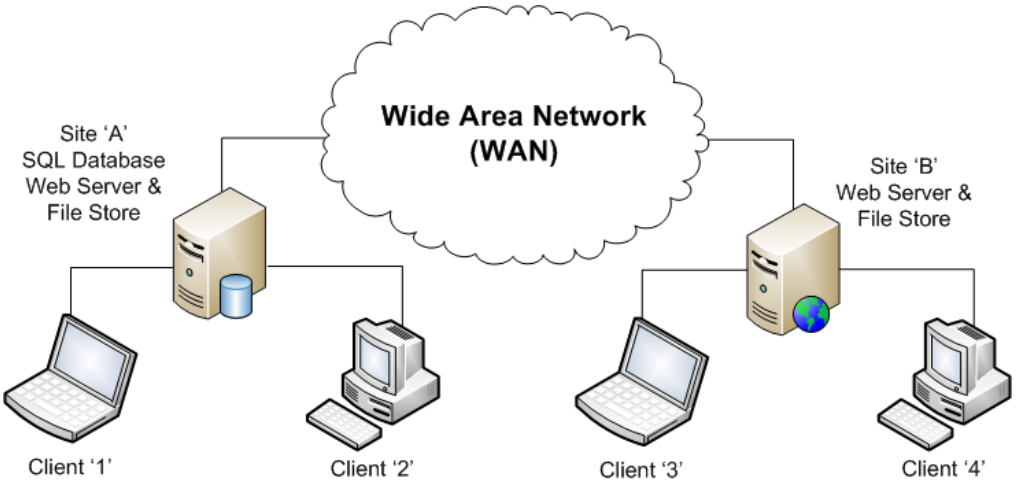
What is a Multiple Site Environment

The multiple site environment allows users to design products across multiple locations as if all of the users were in one location. Autodesk® Productstream® 2009 allows 2 different methods to accommodate this.

The first method is to centralize the data base for all sites. The database for Productstream can sit on a separate server as illustrated below.



The second method is to set up a main site with the database and main filestore located on the same server. The main site should be the site with the most users and where the most design work is being done.



In these environments users open data files such as AutoCAD® drawings Inventor Assemblies or Microsoft® Office documents from there local file Store and all meta data is read from and written to a central database. This allows users to open large files much faster then trying to copy them across the Wide Area Network (WAN)

NOTE Setting up the main site first and attempting to access the main set remotely may provide satisfactory performance for some users. It is recommended to set up the main site and expanding to a multiple site after.

Is Multiple Site Environment Right for You

Setting up a multiple site environment is more complicated than a single server environment and several factors should be taken in consideration when deciding if you want to set up a multiple site environment. You should consider the following:

- Network reliability
- Network latency and Bandwidth between sites. (Should be low latency and high bandwidth.)

- Number of remote sites.
- Number of users per site.
- Size and amount of files to be replicated per day/per hour
- Estimated number of concurrent users

NOTE For more detailed information see the white paper on Productstream multiple site guide lines at *www.autodesk.com/vault-whitepapers* .

Pre-Install Checklist

If you have read the information above and have decided to set up a multiple site environment then before you get started go through the following check list.

- Do all remote sites have the software? Is the software copied on the server? Is there a DVD drive on server?
- Do you have access to an Administrator for all sites? Do you have remote login from main site to each remote sites?
- Do all remote sites have a latest copy of the filestore via a backup package (for existing ADMS customers)?
- Are all Windows updates current on all servers?
- Are virus scans turned off?
- Are firewalls turned off?
- Are TCP Ports 1433 and 1434 blocked by firewall?

Install ADMS on the First Site

The first site in a multiple site environment can be an existing site with Autodesk data management server 2009 already installed or a new installation. If the first site already has Autodesk data management server 2009 installed, proceed to Configure Microsoft SQL Server 2005 for Remote Access on page 78.

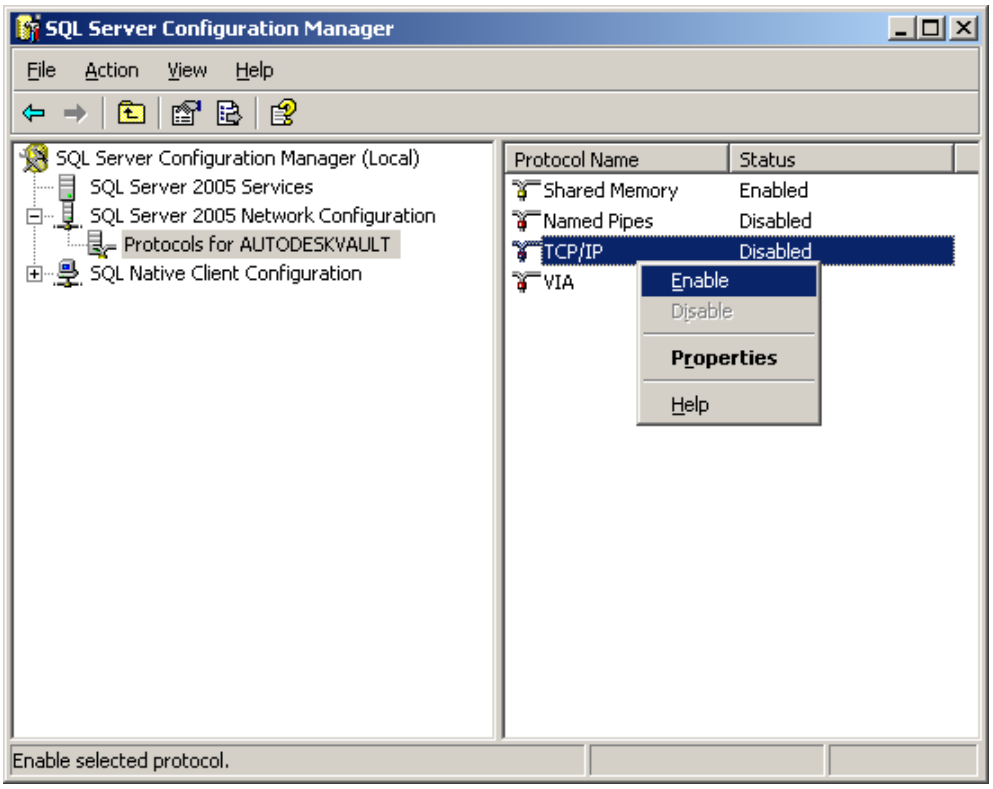
NOTE In a multiple site environment, a remote SQL instance is required for remote sites to access.

If you are installing Autodesk Product stream for the first time please refer to Server Installation on page 22

NOTE When setting up a multiple site environment it is recommended to install the data management server on Microsoft® Windows Server 2003

Configure Microsoft SQL Server 2005 for Remote Access

1. On the computer on which SQL is installed, click Start ➤ Programs ➤ Microsoft® SQL Server 2005 ➤ Configuration Tools ➤ SQL Server Configuration Manager.
2. In the SQL Server Configuration Manager dialog box, expand SQL Server 2005 Network Configuration and then select Protocols for AUTODESKVAULT.
3. Right-click TCP/IP in the protocols list and then select Enable.

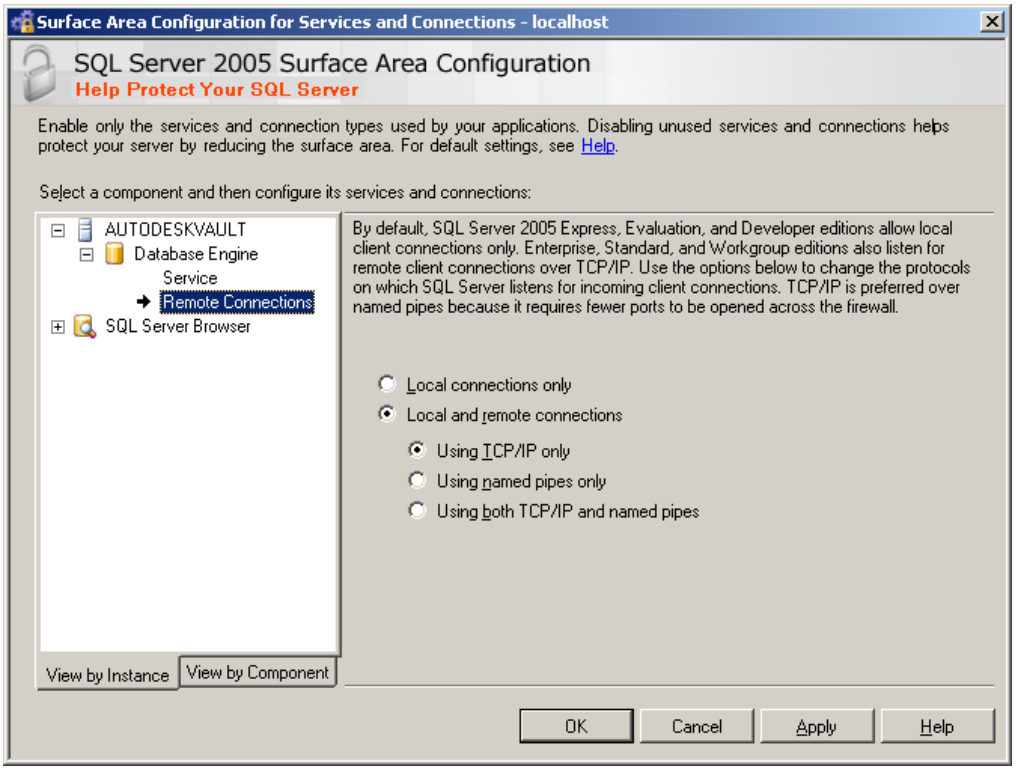


4 Close SQL Server Configuration Manager.

5 On the computer on which SQL is installed, click Start ➤ Programs ➤ Microsoft SQL Server 2005 ➤ Configuration Tools ➤ SQL Server Surface Area Configuration.

6 In the SQL Server Surface Area Configuration dialog box, select Surface Area Configuration for Services and Connections.

7 In the Surface Area Configuration for Services and Connections dialog box, expand AUTODESKVAULT ➤ Database Engine and then select Remote Connections. Verify that Local and Remote Connections and Using TCP/IP only are turned on.



8 Click OK.

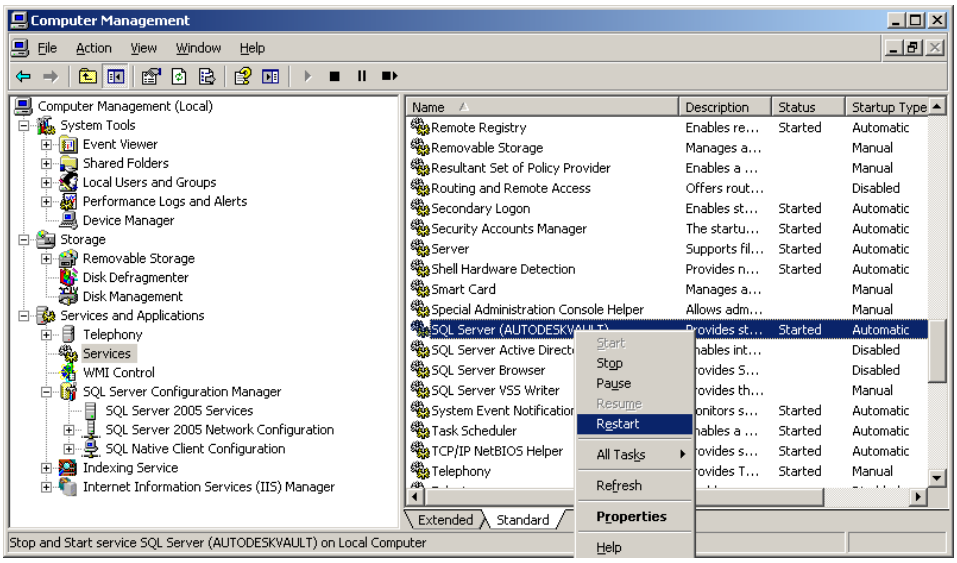
9 From the Windows® Start menu, select Settings ➤ Control Panel.

10 In the Control Panel, double-click Administrative Tools.

11 In the Administrative Tools window, double-click Computer Management.

12 In the Computer Management window, expand Services and Applications and then select Services.

13 Locate the SQL Server (AUTODESKVAULT) service. Right-click the service and then select Restart.



14 Locate the SQL Sever Browser service. Right-click the service and then select Properties.

15 In the SQL Server Browser Properties (Local Computer) dialog box, change the startup type to be Automatic and then click OK.

16 Right-click SQL Server Browser and then select Start.

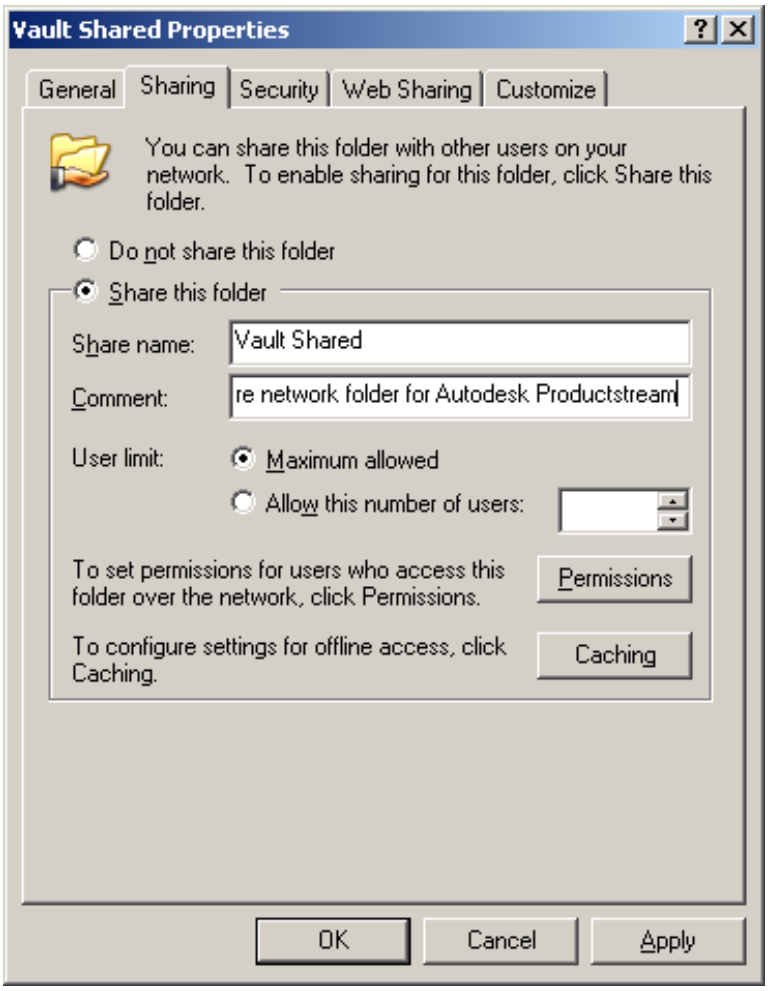
Create Shared Network Folders

In a multi-site environment, one site may host the AUTODESKVAULT SQL instance and the additional sites access that SQL instance remotely, or all sites can access the same SQL instance remotely. Before configuring the additional sites to access the SQL instance, a shared network folder must be created.

The shared network folder is required as a transition area between the SQL instance and the Autodesk data management server. The shared network folder can be located on any computer that is accessible by both the data management server and SQL. The space requirement for the shared folder is equal to the total of all database files.

Both the user account under which the Autodesk data management server is running as well as the user account under which the SQL instance is operating need full access to the shared network folder.

1. Choose a network location and create a new folder.
2. Right-click the new folder and then select Sharing and Security.
3. In the Properties dialog box, click Share this folder.
4. Enter a name for the shared folder.
5. In the Comment field, enter *Shared network folder for Autodesk Productstream*.



6 Click Permissions.

7 In the Permissions dialog box, click Add. Add the user account under which ADMS is running. By default, this is a local administrator account on the ADMS server called Autodesk Vault.

8 Add the Windows account under which the SQL Server service is running. The default is Network Service.

9 Give both accounts Full Control.

10 In the Permissions dialog box, click OK.

11 In the Properties dialog box, click OK.

Install ADMS on Additional Sites

The additional sites in the multi-site environment must have the Productstream data management server installed and configured to access the SQL instance used by the first site. The connection to the remote SQL instance is configured during the installation of the data management server. Follow these steps for each additional site.

1 On the remote site, insert the Autodesk Productstream 2009 installation media.

2 On the installation screen, select either Autodesk Productstream data management server.

3 From the Select the Products to Install page, select Autodesk data management server 2009 and then click Next.

NOTE You can install the data management clients on the server as well, but it is not a requirement.

4 Accept the license agreement and then click Next.

5 On the Review - Configure - Install page, select Autodesk data management server 2009 from the drop-down list and then click Configure.

6 Select the Autodesk data management server 2009 tab.

7 Select Use remote SQL.

Enter the name of the server containing the SQL instance to which to connect or click Browse to locate the server on the network.

NOTE Do not include \\ in front of the server name. A SQL instance named 'AUTODESKVAULT' must exist at the specified location.

8 When using a remote SQL instance, a shared network folder is required as a transition area between the SQL instance and the Autodesk data management server. The shared network folder can be located anywhere on the network and must be accessible by both the data management server and SQL. The space requirement for the shared folder is equal to the total of all database files. Choose a location with sufficient space and performance. Enter the UNC path to the shared network folder or click Browse to locate the shared folder on the network.

NOTE Both the user account under which the Autodesk data management server is running as well as the user account under which the SQL instance is operating need full control access to the shared network folder.

9 Click Configuration Complete.

10 Click Install.

11 The Installation Complete page indicates that Autodesk data management server has been installed. Click Finish.

NOTE Once the server is installed, repeat this process for each additional site.

Managing Sites

Sites are managed and maintained using the server console. For more information and additional site management features, see the server console Help.

1 On the Windows Start menu, select Programs ➤ Autodesk ➤ Autodesk Data Management ➤ Autodesk data management server console.

2 Log into the server console.

3 In the navigation pane, select Sites.

Each site is listed along with a summary of site statistics including:

- Server Name
- Description
- Location of File Store
- Vault Name
- Vault Status
- Replication Type
- Last Replication
- Next Replication
- Site Status

Change Descriptions of Sites

1. Select a site from the list.
2. Right-click the Description field for the site and then select Edit Description.
3. In the Description dialog box, enter a new description for the site and then click OK.

Replicate Data

Replicating data at a site compares the local file store to the file store at each other site and retrieves any missing files from the other sites. Once replicated, a file and its history are available at the replicated site. Individual vaults can be replicated, or all of the enabled vaults on a site can be replicated. Replication can also be scheduled.

NOTE The Replication command synchronizes data only at the site at which it is initiated. Each site is responsible for its own replication.

1. Select the current site from the list.

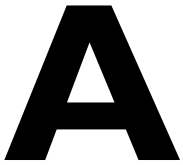
2 Right-click the site, and the select Replication:

- Site Now to synchronize all of the vaults on the current site immediately.
- Vault Now to synchronize only the selected vault.
- Vault Schedule to establish a synchronization schedule or to turn off scheduled synchronization.

3 If you selected Vault Schedule, choose either a daily schedule or a replication frequency in the Scheduled Replication dialog box.

4
- Daily – The default daily replication schedule is weekdays at midnight. Specify a start time and select which days to perform the replication.
- Frequency – Specify a start time and then select how often the replication is performed.
- Click OK.

Performance Tuning



There are several ways to enhance the overall performance of Autodesk® Vault and Productstream®. Before proceeding, it is recommended that you read Understanding Vault Architecture on page 4 in this guide.

Key Terms

Term	Definition
Performance Tuning	Monitoring and analyzing the performance of a system and adjusting its configuration to obtain optimum performance.
Operating System	The software that the rest of the software depends on to make the computer functional. On most PCs this is Microsoft® Windows® or the Mac OS®.
Memory	It is a set of storage locations on PC's main circuit board.
Disk Defragmentation	A utility included with the Windows operating system that coalesces files on the hard drive, freeing up disk space.
Anti-virus Program	A program that searches for viruses and removes any virus that it finds.
Proxy Server	Existing between a client Web browser and a real server, a computer that caches common and often requested Web pages and holds them for easier access by users.

Term	Definition
Customize View	A command in Productstream or Vault Explorer to customize the information shown in the display grid.

Tuning Server Performance

There are a number of factors that can be tuned to get the optimal performance from Autodesk data management server. By reviewing this section and tuning your configuration accordingly, you can get the most from your data management solution.

Operating System

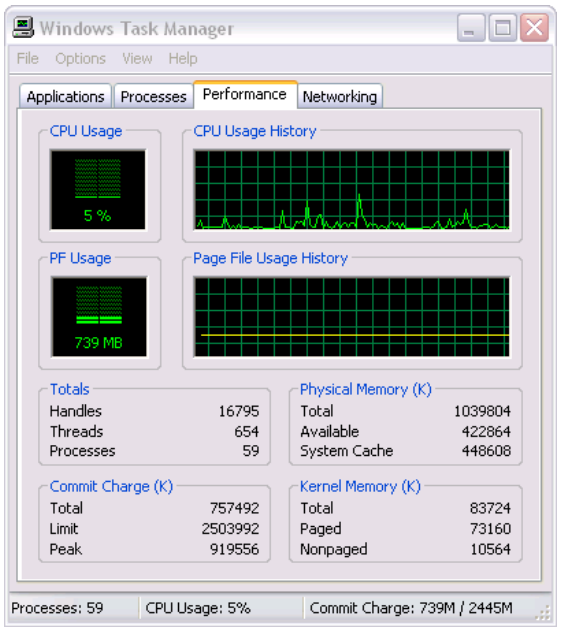
A dedicated server operating system such as Windows server™ 2003 is preferable to workstation operating systems such as Windows XP Professional or multi-purpose server operating systems such as Microsoft Windows Small Business Server 2003.

Server Memory

In order to determine whether the server has enough available memory, some key factors must be considered: the size of your data and the number of users concurrently accessing the Vault server. If Productstream is being used, the number of users concurrently accessing the Item Master should be considered as well.

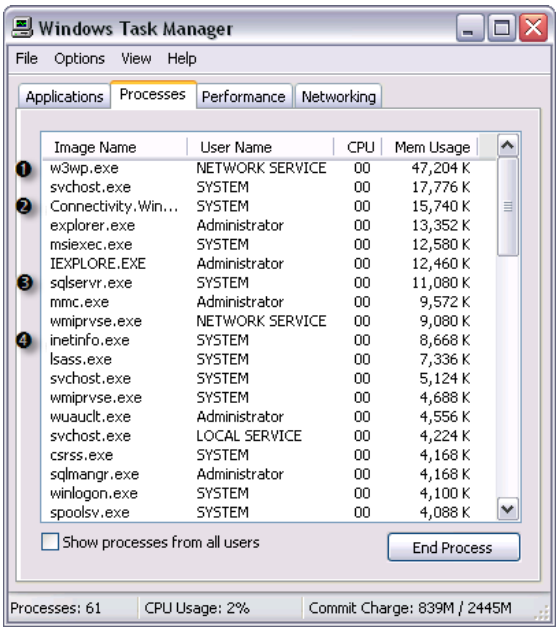
View memory usage

- Invoke the Task Manager by right-clicking in the taskbar or pressing CTRL+ALT+DEL.



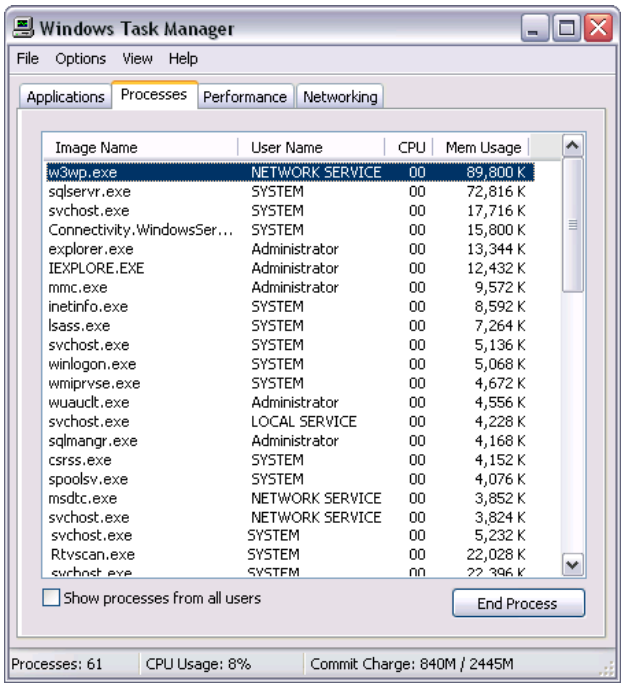
If the Commit Charge Total regularly exceeds the total amount of physical memory, it is likely that many of the programs and processes on the server are using the disk-based swap file excessively. Using the swap file is much slower than using physical memory. Windows uses the swap file for many tasks, so a small amount of swap file usage will not cause any problems.

- If you suspect that your server has an inadequate amount of memory, check the Commit Charge Total value. Clicking the Processes tab will reveal more details on the memory usage of all programs or processes.



Initial memory usage by Autodesk Data Management Server.

1. IIS Web Server process (there may be many of these running)
2. Autodesk data management server notification process
3. Microsoft®SQL server process
4. IIS main Web server process



Memory usage after adding a large Inventor drawing and associated files.

In general, SQL Server memory will grow over time. This is because the database server is using memory as fast cache buffers for frequently used queries. As other programs request memory on the server, SQL Server releases memory as needed. Similarly, the IIS worker process (*w3wp.exe* in Windows 2003 server) reserves memory for caching as well. When the system starts running low on physical memory, the operating system reclaims some of this memory. In general, unless you have a large site with many users or large models, this will not need to be changed.

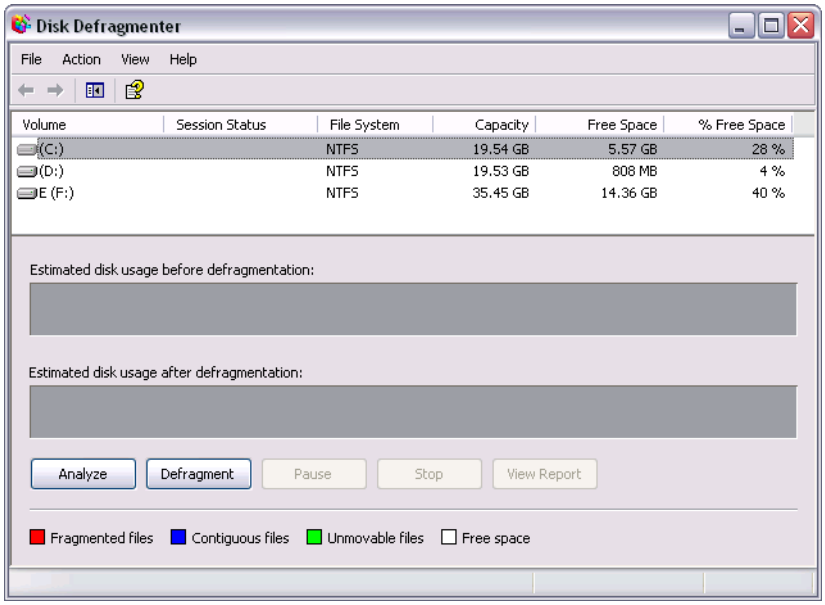
Disk Performance

Using multiple disks will generally enhance system performance. There are many types of disk technologies available, such as the Redundant Array of Inexpensive Disks (RAID). Choosing which type of disk technologies depends on the number of users, vault size, frequency of updates, and so on.

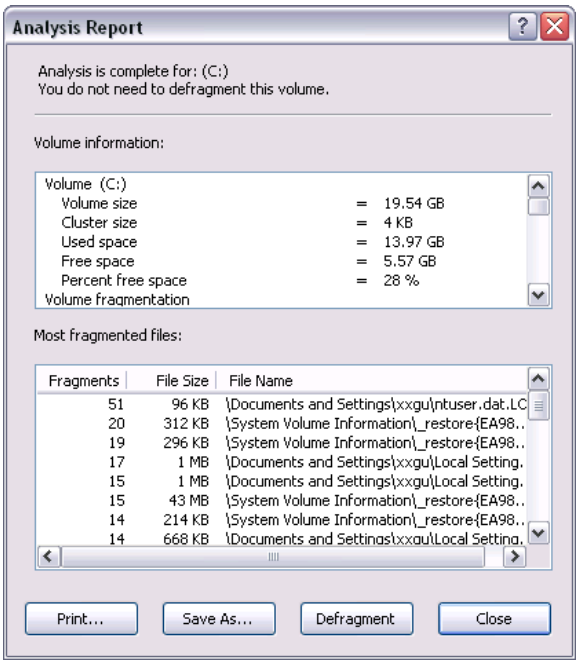
Disk fragmentation can also greatly affect Autodesk data management server performance. Most Windows editions provide a disk tool that measures how the files on your computer are stored on disk.

Running the disk defragmenter

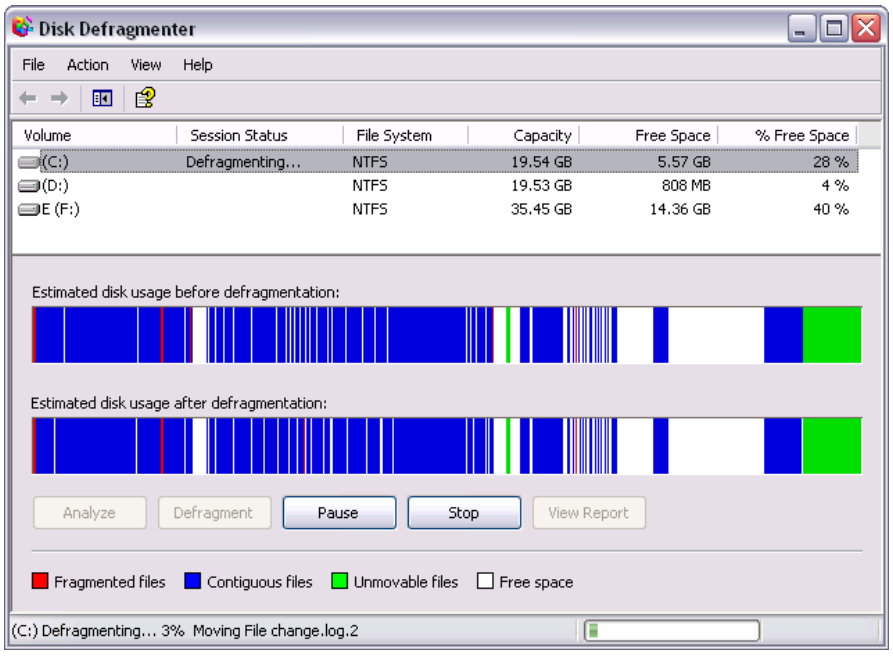
1 Click Start ➤ All Programs ➤ Accessories ➤ System Tools ➤ Disk Defragmenter.



2 Choose a disk (volume) and click Analyze. Following analysis, a report appears.



3 Click the Defragment button.



Aggressive Anti-virus Programs

Some types of Anti-virus programs scan data coming from the network and data being written to the file store. If your anti-virus software appears to be engaged for too long on vault operations, such as uploads, adjusting the scanner settings may help.

There is, of course, a trade-off between security and performance. Because of this, performing some test cases and benchmarks on your own data and server is recommended.

CPU

Faster or multiple CPUs can improve vault performance, but this is typically a more expensive option than the others and should mainly be considered after the other areas have been reviewed.

Vault Client Performance

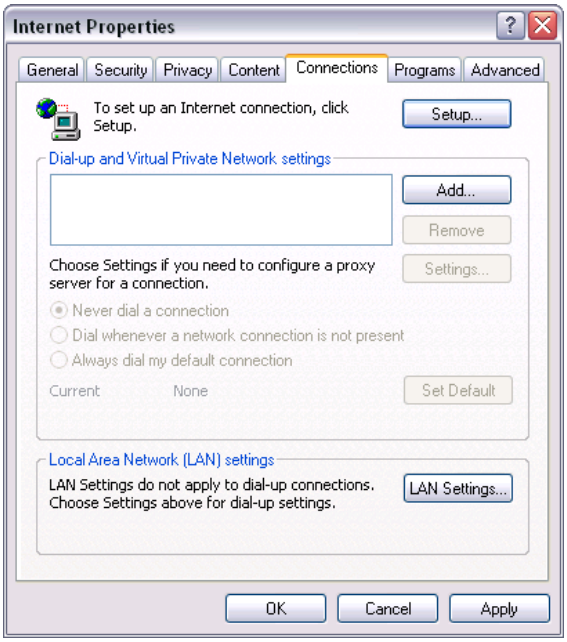
There are a number of factors that can be tuned to get the optimal performance from the data management clients. By reviewing this section and tuning your configuration accordingly, you can get the most from your data management solution.

Network Configuration

Autodesk data management server generally performs better when clients are directly connected to them. Certain types of web setups which use proxy servers or other similar Internet setups are usually of limited value in intranet setups. Bypassing the proxy server for addresses inside your corporate network is therefore recommended.

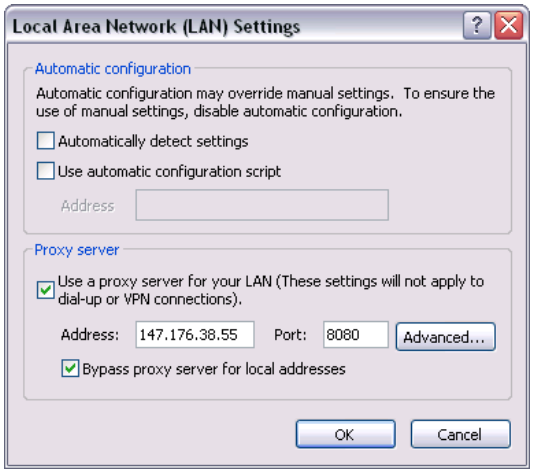
Customize Internet options in Windows Control Panel

1 Click Start ➤ Control Panel, and then double-click Internet Options. In the Internet Properties dialog box, click the Connections tab.



2 Click the LAN Settings button.

3 In the Local Area Network (LAN) Settings dialog box, if the option Use a proxy server for your LAN is selected, it means a proxy server is located in your environment. Consider enabling the Bypass proxy server for local addresses by clicking this option.

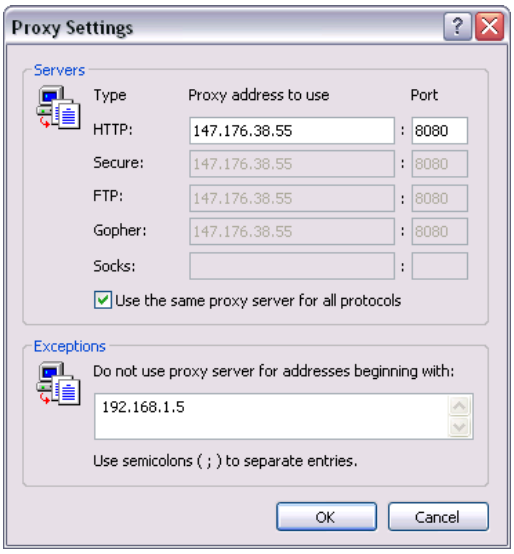


You can also explicitly enter your server's Internet Protocol (IP) address in the bypass column.

Specify the server Internet Protocol (IP) address

1 In the Local Area Network (LAN) Settings dialog box, if a proxy server is specified in the Proxy server section. Click the Advanced button.

2 In the Proxy Settings dialog box enter your vault server address in the Exceptions field.



3 Click OK.

Client Settings

Certain settings within the client application can affect performance not only in the client, but in the Vault add-ins as well.

Customize View command

- This is a commonly used setting in the Productstream and Vault Explorer client programs. It is used to add additional information to display grids. Usually, this will not have much additional impact on performance. However, some display fields, such as the thumbnail property, may take a longer to display. Enabling these settings only on an 'as needed' basis can help to improve day-to-day performance.

Memory

Add-ins for design applications will consume additional memory. They amount of memory they consume is typically small. But, in certain circumstances, it makes sense to review the memory usage on the client. The most common

problems occur when the original client CAD model uses the majority of the computer's available memory before introducing a vault into the workflow. Please refer to the Server Memory on page 88 section for additional instructions on how to review the memory usage on your systems.

Disk Performance

Like the server, client computers can suffer from disk fragmentation as well. The concept of analyzing disk fragmentation is the same for both clients and servers.

Troubleshooting

B

Learn how to use the Autodesk® Server Diagnostics Tool or the Autodesk online Knowledge Base to solve problems that might occur when installing or using Autodesk® data management server.

Key Terms

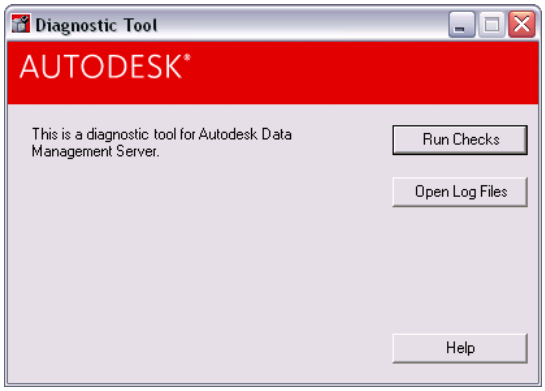
Key Terms	Definition
Autodesk Server Diagnostics Tool	A program that can performs a series of checks to identify possible Autodesk data management server problems and gather system information to help diagnose the problem.
Autodesk Knowledge Base	An online support for all Autodesk products.
Autodesk data management server	The server used in conjunction with Autodesk® Vault, Autodesk® Productstream®, and Content Center for storing files and version data.
Log file	A text file generated by the Autodesk Server Diagnostics Tool containing important system configuration information.

Server Diagnostics Tool

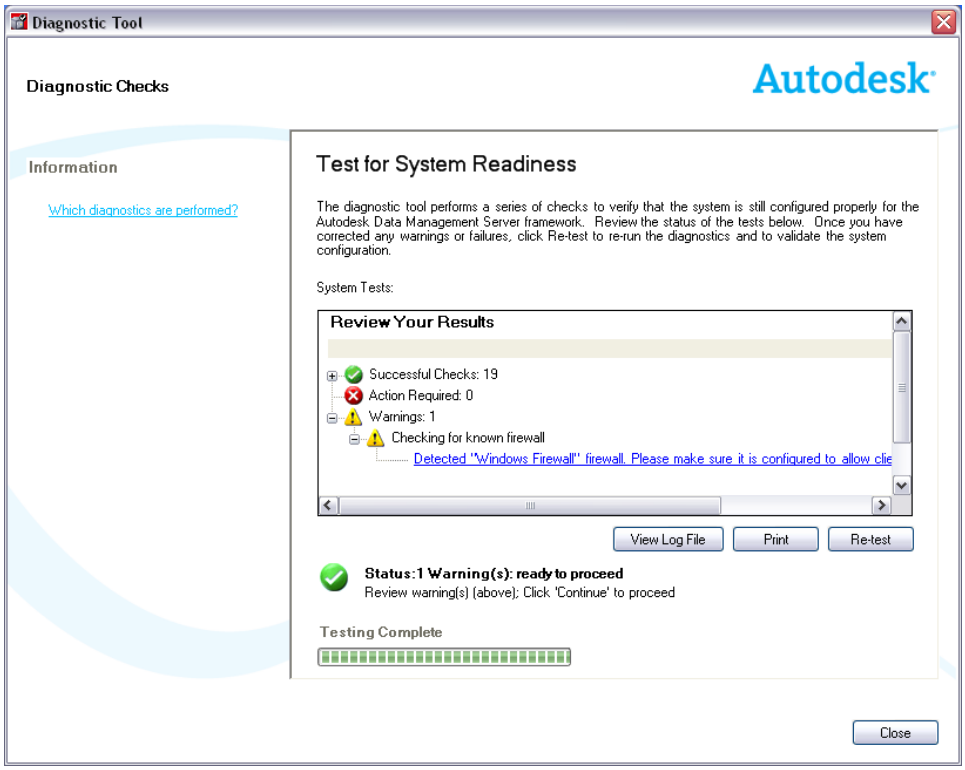
The Autodesk Server Diagnostics Tool is a collection of tests that can help determine why a vault installation is not working properly. Changes to the server machine configuration over time can interfere with the Autodesk data management server. The Autodesk Server Diagnostics Tool performs a series of tests to identify possible Autodesk data management server problems and compiles system information to help diagnose the problem.

Run the diagnostics tool

1. On the Windows® Start menu, select Programs ➤ Autodesk ➤ Autodesk Data Management ➤ Tools ➤ Autodesk Server Diagnostics Tool.
2. In the Diagnostics Tool Setup dialog box, click Run checks.



When Checking, all the results of the diagnostic tests are displayed in a list indicating which of the checks have passed successfully, have generated a warning, or have failed.



- If a warning is returned, click the link in the Warning list to get more information.
- If any of the checks fail, click the link displayed under the Action Required list to learn more about the error.

3 When the diagnostics are complete, click Close and then click Open Log Files to open the directory containing the log files.

Learn more about errors and corrections

- In the Review Your Results list, click the error information under the Action Required list. It is linked to the corresponding pages of Autodesk data management server installation help.

View Diagnostic Log Files

The diagnostic log file lists the name of each condition that is checked along with its information and test results. This includes the version of the operating system, the IIS information, the firewall and any other related configuration information. Other log files are also created during the installation. The log files are stored in the EDMLog folder located in C:\Documents and Settings\Administrator\Local Settings\Temp

Access the log files using one of the following methods:

- In the Diagnostics Tool dialog box, click Open Log File. The EDMLog directory containing all of the log files is opened. Select a log file to open.
- Once you run the diagnostic checks, click View Log File to display the diagnostic log file.

Should it be necessary to seek assistance, you can forward log files to your IT personnel or to Autodesk Technical Support.

Recheck Operating System Environments

After you correct all the errors according to the log files or online help, click Recheck. The Diagnostics Tool will again check the operating system. Repeat this process until the all errors are corrected.

Common Installation Solutions

The following solutions given for installation problems which may occur if the data managment server is not properly configured.

Error 404: File Not Found

During the pre-checks of the data managment server installation you recieve the message “Error 404: File Not Found”.

Solution

This error may occur if the ASP.NET Web servic Extension is not allowed to run in Internet Information Services (IIS). Follow the steps below to resolve the error.

1. Open the Internet Information Services Manager.
2. Expand the Web Services Extension Folder
3. From the right pane, select ASP .NET v2.0.50727.
4. Select the ASP .NET extension in the right pane.
5. Click Allow.
6. Restart the pre-checks.

Error: ASPX failed to run...

During the pre-checks of the data managment server installation you receive the message "Error ASPX failed to run ASP .NET filter is prohibited".

Solution

1. Open the Internet Information Services Manager.
2. Expand the Web Services Extension Folder.
3. From the right pane, select ASP .NET v2.0.50727.
4. Select the ASP .NET extension in the right pane.
5. Click Allow.
6. Restart the pre-checks.

Error: ASPX file failed to run...

During the pre-checks of the data managment server installation you receive the message "ASPX failed to run at URL:'*http://localhost/TestASPNet.aspx*'".

This error may be caused if .NET 2.0 is installed prior to installing Internet Information Services (IIS) and ASP >NET account does not have the proper permissions on the IIS Metabase:

Solution

1. Open a Windows® command prompt and navigate to the *%systemroot%\microsoft.net\framework\2.0.50727* directory.
2. Enter *aspnet_regiis.exe -ga ASPNet* If you are running Windows® XP or Windows Vista™. If you are running Windows server™ 2003 enter *aspnet_regiis.exe -ga "%computername%\Network Service"*
3. Enter *IISRESET*
4. Restart the pre-checks

Error 29001. Performance counter...

During the pre-checks of the data management server installation you receive the message "Error 29001. Performance counter installation error. Input string was not a correct format.".

Solution

This error may occur when the installation attempts to add performance counters for the data management server.

To check the current state of the performance counters

1. Click Start (Windows) ➤ Run.
2. In the Run dialog box, enter *Perfmon*.
3. Verify that you can add performance counters to the list to be monitored.

An inability to add performance counters indicates that system files have been corrupted.

To repair these files

1. Using Windows Internet Explorer®, browse to C:\WINDOWS\system32

2 Rename the perfc009.dat and perfh009.dat files to *perfc009.dat.OLD* and *perfh009.dat.OLD*, respectively.

3 Click Start (Windows) ➤ Run

4 In the Run dialog box, enter *lodctr /r:PerfStringBackup.ini*

5 Export a copy of HKEY_LOCAL_MACHINE\Software\Microsoft\Windows NT\Current Version\Perflib\009 from the system registry of a working computer and transfer it to your computer.

6 Change the Data Values of the Last Counter and Last Help keys so that they are the same as in the system registry of the working computer.

7 Restart your computer.

ISAPI Filter error while running...

While running the data management server on the 64-bit version of Windows server 2003 you notice an Event ID 2274 error in the Event Viewer Application log, indicating a problem with ISAPI filter.

Solution

Refer to the following Microsoft Knowledge Base article:*http://support.microsoft.com/kb/894435/en-us*

Advanced Troubleshooting

Additional troubleshooting resources for the Autodesk data management system can be found online. They include a searchable Knowledge Base available on the Autodesk support page. You can access the Autodesk Knowledge Base online at *www.autodesk.com*.

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